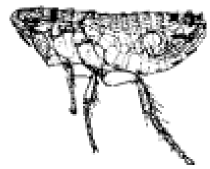
flea



NEWS 83 *December 2018*

http://esanetworks.org/

FLEA NEWS is a twice-yearly newsletter about fleas (Siphonaptera). Recipients are urged to check any citations given here before including them in publications. Many of our sources are abstracting journals and current literature sources, and citations have not necessarily been checked for accuracy or consistent formatting.

Recipients are urged to contribute items of interest to the profession for inclusion herein, including: Flea research citations from journals that are not indexed, Announcements and Requests for material, Contact information for a Directory of Siphonapterists (name, mailing address, email address, and areas of interest - Systematics, Ecology, Control, etc.), Abstracts of research planned or in progress, Book and Literature Reviews, Biography, Hypotheses, and Anecdotes. Send to:

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Editorial

Dear *Flea News* Reader,

The entomologist E.O. Wilson proposes that half the Earth's surface be set aside for preserving biodiversity. Such a policy would preserve an estimated 85% of the world's species, particularly if hotspots were encompassed.

Readers of Flea News already know the significance of fleas and other parasites in biodiversity. The idea of the 'keystone parasite' is gaining acceptance.

In order to diminish extinction rates, we need to employ a variety of approaches, including preserving:

- half the Earth's surface as a refuge for wildlife;
- overall species diversity;
- charismatic species, that can serve as 'umbrellas' for other species;
- keystone species;
- ecosystem functions and services, such as productivity and water purification;
- representative biomes, ecoregions, and special habitats such as caves;
- extreme cases of endangered species in temporary, artificial habitats such as zoos, aquaria, hatcheries, and their associated biotechnology, that can act as buffers to population fluctuations.

Currently, only 15% of terrestrial surface area is protected, still less for marine environments. These percentages are increasing, but so is the pressure to destroy the areas for development.

Yours in fleas,
R.L. Bossard, Editor, *Flea News*

References

Galloway, T. D. (2018). Biodiversity of Ectoparasites: Lice (Phthiraptera) and Fleas (Siphonaptera). Insect Biodiversity: Science and Society, 2, 457-482.

Kwak, M.L. (2018). Australia's vanishing fleas (Insecta: Siphonaptera): a case study in methods for the assessment and conservation of threatened flea species. Journal of Insect Conservation, 22 (3-4), 545-550.

Wilson, E. O. (2016). Half-Earth: Our Planet's Fight for Life. Liveright Publ., New York.

Announcements

AMERICAN ASSOCIATION OF VETERINARY PARASITOLOGISTS, WORLD ASSOCIATION FOR THE ADVANCEMENT OF VETERINARY PARASITOLOGY, and Livestock Insect Workers Conference in Madison, Wisconsin (July 7-11, 2019).

Registration is open, as well as housing and abstract submission. http://www.waavp2019.com/ The LIWC Lifetime Achievement and Industry awards are also on the website and taking submissions.

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Flea-related Abstracts
Nov. 2018 ESA, ESC, and ESBC Joint Annual Meeting
Vancouver, British Columbia

Optimizing the off-host production of cat fleas, *Ctenocephalides felis*, using a membrane-based feeding system.

Brittny Blakely (brittnyb@nmsu.edu) and Alvaro Romero, New Mexico State Univ., Las Cruces, NM

Cat fleas, a common blood feeding parasite known to feed on humans and domesticated animals worldwide, cause annoyance and discomfort through their bites, and can transmit several diseases including the plague, murine typhus and cat scratch disease. Cat fleas reared in laboratory conditions are needed to study disease transmission, develop new anti-flea formulations, and monitor insecticide resistance. Traditionally, fleas have been reared on live animals, but this requires animal handling permits, inflicts discomfort to the animals, and host animal upkeep can be costly and time consuming. An alternative is the use of an artificial membrane-based feeding system, however this method may not be sustainable long term, as fleas reared off-host produce fewer eggs than traditionally reared fleas. This study refines previously described methods of rearing cat fleas using artificial membrane-based feeders. We initially compared blood consumption, adult mortality, egg production, and hatching rates of fleas fed defibrinated or sodium citrated bovine blood at 40°, 37°, or 34° Celsius. Results show that cat fleas consume more defibrinated bovine blood offered at 40°C or 37°C, and have lower adult mortality compared to other blood and temperature combinations. Compared to previous studies, our egg production seems to be higher, but hatching rate is lower. We will further improve reproductive and

developmental variables by evaluating the effects of adult feeding duration and parental sex ratios. Developing a reliable and convenient method to rear adequate cat fleas without live animals will allow more for more humane and cost-efficient research on this important pest.

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Identification of fleas on sciurid rodents in southern Saskatchewan using PCR-based techniques.

Jessica Thoroughgood (jessica.thoroughgood@usask.ca) and Neil Chilton, Univ. of Saskatchewan, Saskatoon, SK, Canada

Fleas are vectors of bacterial pathogens transmitted to vertebrate hosts. Some species, such as *Oropsylla rupestris*, are vectors of *Yersinia pestis*, the causative agent of plague. It is important to distinguish these flea species from those not capable of transmitting *Y. pestis* in order to determine the potential risk of infection within rodent populations. Fleas are identified based on their morphological characteristics. This requires chemical clearing of the internal structures, a process that takes several weeks, and which prevents subsequent DNA studies of the fleas and their associated bacteria. Molecular techniques provide a valuable alternative for the identification of fleas and bacteria within their microbiomes. In this study, several species of flea, including *O. rupestris*, collected from three species of sciurid rodents, *Urocitellus richardsonii* (Richardson's ground squirrels), *Ictidomys tridecemlineatus* (13-lined ground squirrels), and *Cynomys ludovicianus* (black-tailed prairie dogs) in southern Saskatchewan, were identified using PCR-based techniques. The four potential genetic markers examined were the nuclear 28S rRNA and 18S rRNA genes, and two mitochondrial cytochrome oxidase (cox) genes. All, but the 18S rRNA gene, were useful for delineating among species, and for inferring phylogenetic relationships. In addition, intraspecific variation was detected for the cox genes suggesting that these genetic markers may be useful for population genetic studies.

[Editor's note: clearing of fleas takes days usually, not weeks.]

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Vector-borne pathogens in fleas from free-ranging *Cerdocyon thous* (crab-eating fox) in southern Brazil.

Diogo Schott (diogo.schott@yahoo.com) et al. Instituto de Pesquisas Veterinárias Desidério Finamor, Eldorado do Sul, Brazil, Área de Vida Assessoria e Consultoria em Biologia e Meio Ambiente, Canoas, Brazil, Ka'aguy Consultoria Ambiental, Pelotas, Brazil, Univ. Paul Sabatier, Porto Alegre, Brazil, Fundação Zoobotânica do Rio Grande do Sul (FZB-RS), Porto Alegre, Brazil

Fleas are worldwide distributed insects that have been implicated in the transmission of several pathogens. The aim of this work is to investigate the presence of the vector-borne pathogens *Rickettsia* spp. and *Bartonella* spp. in fleas from free-ranging crab-eating foxes (*Cerdocyon thous*) from Rio Grande do Sul State, Southern Brazil. Foxes were captured in six municipalities. Fleas were manually collected from animals, immediately placed in ethanol, and identified based on a dichotomous morphological key. For molecular detection of *Rickettsia* spp. and *Bartonella* spp., the total DNA of the collected fleas was extracted individually and submitted for PCR. Twenty-nine C. thous were sampled and four were parasitized by fleas, all of which were identified as *Ctenocephalides felis*. Ten fleas were collected from these animals — eight of them positive for *Rickettsia* spp., three positive for *R. felis*, and five positive for *R. asembonensis*. Regarding *Bartonella* spp., 40% (4/10) showed positive results in the PCR. The areas in which *C. felis* were collected show different environmental conditions, including two different biomes. The set of results presented here show that the cosmopolitan invasive cat flea (*C. felis*) and its related flea-borne pathogens are found in association with free-ranging Neotropical wild canids.

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Biofilm blockage of the flea's foregut mediates flea-borne plague transmission.

Viveka Vadyvaloo (vvadyvaloo@ wsu.edu), Washington State Univ., Pullman, WA

[abstract not available]

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Thanks to everyone sending corrections to the Updated Listing of Flea Species of the World. R.L. Bossard, Editor, *Flea News*

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Featured Research

New Flea Species 2018

Sanchez, J., Lareschi, M., Salazar-Bravo, J., Gardner, S.L.

Fleas of the genus *Neotyphloceras* associated with rodents from Bolivia: new host and distributional records, description of a new species and remarks on the morphology of *Neotyphloceras rosenbergi*.

(2018) Medical and Veterinary Entomology, 32 (4), pp. 462-472.

[*Neotyphloceras boliviensis* n. sp.]

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López-Berrizbeitia, M.F., Sanchez, J., Barquez, R.M., Díaz, M.M.

Descriptions of two new species of flea of the genus *Plocopsylla* in northwestern Argentina

(2018) Medical and Veterinary Entomology, 32 (3), pp. 334-345.

[Family Stephanocircidae. Subfamily Stephanocircinae.]

[*Plocopsylla (Plocopsylla) chicoanaensis* López-Berrizbeitia, Sanchez, Barquez & Díaz sp. n.]
[*Plocopsylla (Plocopsylla) hastriteri* López-Berrizbeitia, Sanchez, Barquez & Díaz sp. n.]

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Taxonomy

Beaucournu AD Zewdneh TT Bereket Laudisoit AA & A. 2018. Description of the male of *Ctenophthalmus (Ethioctenophthalmus) vanhoutteae* Beaucournu & Bereket, 2017. Bulletin of the Entomological Society of France, 123: 329-332.

C. Briand. 2017. Collection of Siphonaptera Beaucournu AD: The Siphonaptera of the Armorican. Invertebrates Armoricans, No. 17, pp.1 - 39.

Zurita, A., Djeghar, R., Callejón, R., Cutillas, C., Parola, P., & Laroche, M. (2018). Matrix-assisted laser desorption/ionization time-of-flight mass spectrometry as a useful tool for the rapid identification of wild flea vectors preserved in alcohol. Medical and Veterinary Entomology, doi:10.1111/mve.12351

Packer, L., Monckton, S. K., Onuferko, T. M., & Ferrari, R. R. (2018). Validating taxonomic identifications in entomological research. Insect Conservation and Diversity, 11(1), 1-12. https://onlinelibrary.wiley.com/doi/pdf/10.1111/icad.12284\

We surveyed the treatment of taxonomic information in 567 papers published in nine entomological journals in 2016. The proportion of papers that provide taxonomic data in sufficient detail to permit precise validation of taxonomic identifications is vanishingly small: most did not cite identification methods, most did not state whether identified material had been vouchered, and taxon concepts were almost universally absent in non-taxonomic papers. Overall, the combination of all three factors was provided less than 2% of the time and almost two-thirds of all papers provided none of the three.

We suggest that journals should modify the templates used by editors and reviewers by overtly including the following questions:

i. Are Order and Family named in the title, abstract or keywords?

ii. Are the methods used for identification of all studied taxa stated clearly?

iii. Is it clear who did the identifications, are they named and is their contact information and/or institutional affiliation provided?

iv. Is the literature whereupon these identifications are based cited appropriately? This would include some reference to as thorough a revisional taxon concept statement as possible, preferably from recent revisions if available.

v. Are exemplars of all focal species (or all sampled individuals) vouchered in a named repository (ideally with contact person name and accession numbers or other means of ready detection)?

Accurate and replicable taxonomic identification is the cornerstone of biology, without which entomological research risks becoming irreproducible and thus not scientific.

Key words. Identification keys, instructions to authors, revisional taxon concept, vouchers.

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Jason L Malaney, Joseph A Cook; A perfect storm for mammalogy: declining sample availability in a period of rapid environmental degradation, Journal of Mammalogy, gyy082, https://doi.org/10.1093/jmammal/gyy082

Natural history collections have stimulated insights into systematics and evolution, but the extensive biodiversity sampling held in museums is increasingly employed to address other critical societal concerns, especially those related to changing environmental conditions on our planet. Due to large-scale digitization efforts in the last decade, specimen information can now be collated across natural history museums. Here, we leverage the availability of digital records of specimens in the United States that span the past ~135 years to explore the vitality of this resource. Using mammals as an example, we document a significant decline in recent specimen acquisition at a time of extreme environmental degradation and loss of mammalian populations. To stimulate rigorous assessments of the impacts of changing conditions and future-proof this basic infrastructure for mammalogy, we recommend a renewed effort to build temporally deep, geographically extensive, and site-intensive collections of holistic specimens. Targeted fieldwork should be designed to leverage historic sampling to enable retrospective environmental analyses and derive more complete perspectives of change.

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Hosts and Distribution

Aguilar-Montiel, F., A. Estrada Torres, R. Acosta, M. Rubio-Godoy, and J. Vázquez. 2018. Host species influence on flea (Siphonaptera) infection parameters of terrestrial micromammals in a temperate forest of Mexico. Parasitology 1–8. https://doi.org/10.1017/S0031182018001981

Studies of abundance and distribution of organisms are fundamental to ecology. The identity of host species is known to be one of the major factors influencing ectoparasitic flea abundance, but explanations are still needed regarding how host taxa influence abundance parameters of different flea species. This study was carried out at La Malinche National Park (LMNP), Tlaxcala, Mexico, where previously 11 flea species had been recorded on 8 host species. Our aims were to list micromammal flea species, to determine flea infection parameters [flea prevalence (FP) and flea mean abundance (FMA)] and to analyse the influence of host species on these parameters. A total of 16 species of fleas were identified from 1178 fleas collected from 14 species of 1274 micromammals captured with Sherman® traps from March 2014 to December 2015 in 18 sites at LMNP. Some host species influence FP and FMA, in particular, *Microtus mexicanus* and *Peromyscus melanotis* showed particularly higher infection values than other host species. *Plusaetis aztecus* and *Plusaetis sibynus* were identified as the most abundant flea species.

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McAllister, C. T., Durden, L. A., Robison, H. W., & Connior, M. B. (2017). The Fleas (Arthropoda: Insecta: Siphonaptera) of Arkansas. Journal of the Arkansas Academy of Science, 71(1), 69-76.

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Galloway, T.D. 2018. Biodiversity of ectoparasites: lice (Phthiraptera) and fleas (Siphonaptera). In Insect biodiversity. Science and Society, 2nd Edition. Edited by R.G. Foottit and P.H. Adler, Wiley-Blackwell Press, pp. 457-482.

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Kwak, M.L. (2018)
Australia's vanishing fleas (Insecta: Siphonaptera): a case study in methods for the assessment and conservation of threatened flea species.Journal of Insect Conservation, 22 (3-4), pp. 545-550.
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Lareschi M. 2018. Description of the males of *Androlaelaps misionalis* and *Androlaelaps ulysespardinasi* (Acari: Parasitiformes: Laelapidae) parasitic of sigmondontine rodents from northeastern Argentina. Journal of Parasitology 104 (4): 372-376. https://doi.org/10.1645/18-1

Lareschi M., JM Venzal, S Nava, AJ Mangold, A Portillo, AM Palomar Urbina & JA Oteo Revuelta. 2018. The human flea *Pulex irritans* Linnaeus, 1758 (Siphonaptera: Pulicidae) and an investigation of *Bartonella* and *Rickettsia* in northwestern Argentina. Revista Mexicana de Biodiversidad 89: 375-381

Sanchez J, Lareschi M. 2018. Ecological approach of the fleas associated with Sigmodontine rodents from Patagonia, Argentina. Bulletin of Entomological Research. doi:10.1017/S0007485318000196

Sanchez J., A. Travaini, A. Rodriguez & M. Lareschi. 2018. Primer estudio de los índices parasitológicos de dos especies de pulgas (Insecta, Siphonaptera) parásitas de zorros de la Patagonia Argentina: *Pulex irritans* (Pulicidae) y *Polygenis (P.) platensis* (Rhopalopsyllidae). Mastozoología Neotropical 25(1):251-255

Liljesthrom G, Lareschi M. Predicting species richness of ectoparasites of wild rodents from the Río de la Plata coastal wetlands, Argentina. Parasitology Research 117: 2507–2520 https://doi.org/10.1007/s00436-018-5940-5

Sanchez J., M. Lareschi, J. Salazar-Bravo & S. L. Gardner. Fleas of genus *Neotyphloceras* associated with rodents from Bolivia: New host and distributional records, description of a new species and remarks on the morphology of *Neotyphloceras rosenbergi*. Medical and Veterinary Entomology 32: 462-472. https://doi.org/10.1111/mve.12314

Arce S., D.E. Manzoli, M. J. Saravia-Pietropaolo, M. A. Quiroga, L. R. Antoniazzi, M. Lareschi; P. M. Beldomenico. The tropical fowl mite, Ornithonyssus bursa (Acari: Macronyssidae): Environmental and host factors associated with its occurrence in Argentine passerine communities. Parasitology Research https://doi.org/10.1007/s00436-018-6025-1

Sanchez J.P., Travaini, A., Rodríguez, A. & M. Lareschi. 2018. Primer Registro de Pulgas (Insecta, Siphonaptera) en poblaciones de *Lycalopex* (Carnivora, Canidae) de la Patagonia Argentina. Mastozoología Neotropical 25(1): 251-255.

Sanchez J.P. & Y. Gurovich. 2018. Fleas (Insecta: Siphonaptera) Associated to the endangered Neotropical marsupial Monito del Monte (*Dromiciops gliroides* Microbiotheria: Microbiotheriidae) in its Southernmost Population of Argentina. Mastozoología Neotropical 25(1): 257-262.

Ecology

Castaño-Vázquez, F., Martínez, J., Merino, S., Lozano, M. (2018)
Experimental manipulation of temperature reduce ectoparasites in nests of blue tits *Cyanistes caeruleus.*
Journal of Avian Biology, 49 (8), art. no. e01695

Several models predict changes in the distributions and incidences of diseases associated with climate change. However, studies that investigate how microclimatic changes may affect host–parasite relationships are scarce. Here, we experimentally increased the temperature in blue tit *Cyanistes caeruleus* nest boxes during their breeding season to determine its effects on the parasitic abundance (i.e. of nest-dwelling ectoparasites, blood-sucking flying insects and hemoparasites) in nests and the host condition of nestlings and adults. The temperature was increased using heat mats placed underneath the nest material, which resulted in an average temperature increase of 3ºC and a reduction in relative humidity of about six units. The abundance of mites *Dermanyssus gallinoides* and blowfly pupae *Protocalliphora azurea* was significantly reduced in heated nest boxes. Although not statistically significant, a lower prevalence of flea larvae *Ceratophyllus gallinae* was also found in heated nests. However, heat treatment did not affect hemoparasite infection of adult blue tits or the body condition of adult and nestling blue tits. In conclusion, heat treatment in blue tit nests reduced nest-dwelling ectoparasites yet without any apparent benefit for the host.

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Boyes, D.H., Lewis, O.T.

Ecology of Lepidoptera associated with bird nests in mid-Wales, UK

(2019) Ecological Entomology, 44 (1), pp. 1-10. DOI: 10.1111/een.12669

Bird nests are ubiquitous but patchy resources in many terrestrial habitats. Nests can support diverse communities of commensal invertebrates, especially moths (Lepidoptera). However, there is a shortage of information on the moths associated with bird nests, and the factors influencing their abundance, diversity and composition.

Two hundred and twenty-four nests, from 16 bird species, were sampled from sites in mid-Wales (UK) and the moths that emerged from them were recorded.

Seventy eight percent of nests produced moths, with 4657 individuals of ten species recorded. Moth communities were dominated by generalist species rather than bird nest specialists.

Open nests built in undergrowth supported significantly fewer moths than nests in enclosed spaces (for example, nesting boxes). The occurrence of fleas was positively associated with the incidence and abundance of moths. There was no evidence that different nest types supported different moth communities.

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do Couto, A. P., Da Silveira, R., Soares, A. V., & Menin, M. (2018). Diet of the Smoky Jungle Frog *Leptodactylus pentadactylus* (Anura, Leptodactylidae) in an urban forest fragment and in a pristine forest in Central Amazonia, Brazil. Herpetology Notes, 11, 519-525.

Understanding the natural history of amphibians in human altered habitats is essential to develop conservation and management actions. In this study, we determined the composition of the diet of *Leptodactylus pentadactylus* in an urban forest fragment and in a preserved forest, both placed in Central Amazonia, Brazil. Diet samples were obtained by stomach flushing of 33 individuals. Each prey item was measured and identified according to its taxonomic group. For each taxon found in the stomachs we determined the number of items, the percentages of volume, frequency and occurrence, and the index of relative importance. We tested for differences in trophic niche breadth, and the relationship between individual size and prey volume. We identified a total of 127 prey items belonging to 18 different taxonomic groups. Arthropods were the main source of food on both areas. Based on the index of relative importance Araneae, Scorpiones, Diplopoda, and Coleoptera were the most important prey items in the forest fragment while Araneae, Diplopoda, Coleoptera, and Diptera were the most important prey items in the preserved forest. Additionally, one small lizard species, *Alopoglossus angulatus,* was consumed by one of the frogs at the forest fragment. There was no significant difference in the trophic niche breath values obtained between the areas, and no correlation between the largest prey items consumed and body sizes of the frog individuals. Overall, the diet of the L. pentadactylus was similar in both sites and follows a generalist and opportunistic pattern resembling other species of medium or large-sized L*eptodactylus*.

[extract: "The occurrence of very small invertebrates in the diet of *L. pentadactylus* such as Acari, Collembola, and Siphonaptera, could be due to accidental ingestion during capture of larger preys in the leaf litter".]

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Nicholas J. Clark, Jennifer M. Seddon, Jan Šlapeta and Konstans Wells. 2018. Parasite spread at the domestic animal - wildlife interface: anthropogenic habitat use, phylogeny and body mass drive risk of cat and dog flea (*Ctenocephalides* spp.) infestation in wild mammals. Parasites & Vectors 11:8 https://doi.org /10.1186/s13071-017-2564-z

Spillover of parasites at the domestic animal - wildlife interface is a pervasive threat to animal health. Cat and dog fleas (*Ctenocephalides felis* and *C. canis*) are among the world's most invasive and economically important ectoparasites. Although both species are presumed to infest a diversity of host species across the globe, knowledge on their distributions in wildlife is poor. We built a global dataset of wild mammal host associations for cat and dog fleas, and used Bayesian hierarchical models to identify traits that predict wildlife infestation probability. We complemented this by calculating functional-phylogenetic host specificity to assess whether fleas are restricted to hosts with similar evolutionary histories, diet or habitat niches.

Over 130 wildlife species have been found to harbour cat fleas, representing nearly 20% of all mammal species sampled for fleas. Phylogenetic models indicate cat fleas are capable of infesting a broad diversity of wild mammal species through ecological fitting. Those that use anthropogenic habitats are at highest risk. Dog fleas, by contrast, have been recorded in 31 mammal species that are primarily restricted to certain phylogenetic clades, including canids, felids and murids. Both flea species are commonly reported infesting mammals that are feral (free-roaming cats and dogs) or introduced (red foxes, black rats and brown rats), suggesting the breakdown of barriers between wildlife and invasive reservoir species will increase spillover at the domestic animal - wildlife interface.

Our empirical evidence shows that cat fleas are incredibly host-generalist, likely exhibiting a host range that is among the broadest of all ectoparasites. Reducing wild species' contact rates with domestic animals across natural and anthropogenic habitats, together with mitigating impacts of invasive reservoir hosts, will be crucial for reducing invasive flea infestations in wild mammals.

Evolution

Medvedev, S.G.
Morphological Diversity of Fleas' Structures (Siphonaptera). Part 5: Features of Thoracic Combs and Legs
(2018) Entomological Review, 98 (7), pp. 819-825.

37 states of 9 characters of the thoracic combs, coxae, and the 5th hind tarsomere of fleas are analyzed. Some of these character states mark groups of families in the infraorders Pulicomorpha, Hystrichopsyllomorpha, Pygiopsyllomorpha and Ceratophyllomorpha.

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Medvedev, S.G.
Morphological Diversity of the Skeletal Structures and Problems of Classification of Fleas (Siphonaptera). Part 6
(2018) Entomological Review, 98 (1), pp. 10-20.

The structure of 76 skeletal elements of adult fleas was analyzed, and the distribution of 114 characters with 446 character states over the body tagmata, segments, and morphofunctional complexes was investigated. Among them, 40% of the characters (40) and their states (163) describe the diversity of the structures of the frontal complex (including those of the head and prothorax), which is related to the specific features of flea parasitism. A large part of the characters (18) and their states (83) describe the structures of the nototrochanteral complex of the meso- and metathorax responsible for jumping. The total number of all types of homoplasies (258 states) is almost 1.8 times as great as the number of the states (145) that may be regarded as synapomorphies. The ancestral states (43) comprise a smaller portion of the total number. The proportion of the synapomorphic and homoplastic character states varies between the morphofunctional complexes.

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De Lima, F.C.G., De Oliveira Porpino, K.
Ectoparasitism and infections in the exoskeletons of large fossil cingulates
(2018) PLoS ONE, 13 (10), art. no. e0205656.

Studies on paleopathological alterations in fossil vertebrates, including damages caused by infections and ectoparasites, are important because they are potential sources of paleoecological information. Analyzing exoskeleton material (isolated osteoderms, carapace and caudal tube fragments) from fossil cingulates of the Brazilian Quaternary Megafauna, we identified damages that were attributed to attacks by fleas and dermic infections. The former were compatible with alterations produced by one species of flea of the genus *Tunga,* which generates well-delimited circular perforations with a patterned distribution along the carapace; the latter were attributable to pathogenic microorganisms, likely bacteria or fungi that removed the ornamentation of osteoderms and, in certain cases, generated craters or pittings. Certain bone alterations observed in this study represent the first record of flea attack and pitting in two species of large glyptodonts (*Panochthus* and *Glyptotherium*) and in a non-glyptodontid large cingulate (*Pachyarmatherium*) from the Quaternary of the Brazilian Intertropical Region. These new occurrences widen the geographic distribution of those diseases during the Cenozoic and provide more evidence for the co-evolutionary interaction between cingulates and parasites registered to date only for a small number of other extinct and extant species.

[Editor's note: Cingulata are placental, armored mammals from the Nearctic and Neotropical, of which only armadillos survive; some were 2,000 kg (4,400 lb)+.]

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Pielowska, A., Sontag, E., Szadziewski, R.
Haematophagous Arthropods in Baltic Amber
(2018) Annales Zoologici, 68 (2), pp. 237-249.

Haematophagous i.e. blood-feeding or blood-sucking arthropods described from Paleogene Baltic amber are reviewed and commented on. Arthropods feeding on blood from mammals and birds, and occasionally on reptiles and amphibians, are reported as inclusions in fossil resins dated back to the Lower Cretaceous. Eocene Baltic amber from deposits in the Gulf of Gdańsk, Rovno and Bitterfeld, dated from 35 to 50 million years ago, contains 48 fossil species of blood-feeding arthropods placed in extant and extinct genera. Haematophagous fossil arthropods from Acari (1 species), Phthiraptera (+), Siphonaptera (4), and Diptera (43 species) are reported in Baltic amber. Blood-sucking flies are represented by six families: Ceratopogonidae (11), Corethrellidae (5), Culicidae (5), Psychodidae (5), Simuliidae (9), and Tabanidae (8 species). The percentage of species of blood-sucking dipterans in the Baltic amber forest was similar to that in the extant fauna of Poland (3.4%, and 3.2%, respectively). A catalogue of named haematophagous arthropods reported from Baltic amber is provided.

[Family: Ctenophthalmidae, Subfamily: Ctenophthalminae]

Palaeopsylla similis Dampf, 1910
Palaeopsylla baltica Beaucournu et Wunderlich, 2001
Palaeopsylla dissimilis Peus, 1968
Palaeopsylla groehni Beaucournu, 2003

Palaeopsylla klebsiana Dampf, 1911
Palaeopsylla sp. Hoffeins et al. 2018

[Editor's note: It's unclear why 4 flea species are listed in abstract but 6 are listed in article, or why Phthiraptera are listed as "+".]

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Appelgren, A.S.C., Saladin, V., Richner, H., Doligez, B., McCoy, K.D.
Gene flow and adaptive potential in a generalist ectoparasite
(2018) BMC Evolutionary Biology, 18 (1), art. no. 99.

Background: In host-parasite systems, relative dispersal rates condition genetic novelty within populations and thus their adaptive potential. Knowledge of host and parasite dispersal rates can therefore help us to understand current interaction patterns in wild populations and why these patterns shift over time and space. For generalist parasites however, estimates of dispersal rates depend on both host range and the considered spatial scale. Here, we assess the relative contribution of these factors by studying the population genetic structure of a common avian ectoparasite, the hen flea *Ceratophyllus gallinae*, exploiting two hosts that are sympatric in our study population, the great tit *Parus major* and the collared flycatcher *Ficedula albicollis*.

Previous experimental studies have indicated that the hen flea is both locally maladapted to great tit populations and composed of subpopulations specialized on the two host species, suggesting limited parasite dispersal in space and among hosts, and a potential interaction between these two structuring factors.

Results: *C. gallinae* fleas were sampled from old nests of the two passerine species in three replicate wood patches and were genotyped at microsatellite markers to assess population genetic structure at different scales (among individuals within a nest, among nests and between host species within a patch and among patches). As expected, significant structure was found at all spatial scales and between host species, supporting the hypothesis of limited dispersal in this parasite. Clustering analyses and estimates of relatedness further suggested that inbreeding regularly occurs within nests. Patterns of isolation by distance within wood patches indicated that flea dispersal likely occurs in a stepwise manner among neighboring nests. From these data, we estimated that gene flow in the hen flea is approximately half that previously described for its great tit hosts.

Conclusion: Our results fall in line with predictions based on observed patterns of adaptation in this host-parasite system, suggesting that parasite dispersal is limited and impacts its adaptive potential with respect to its hosts. More generally, this study sheds light on the complex interaction between parasite gene flow, local adaptation and host specialization within a single host-parasite system.

Pathology and Control

Hufthammer AK, Walløe L (2013) Rats cannot have been intermediate hosts for *Yersinia pestis* during medieval plague epidemics in Northern Europe. J Archaeol Sci 40:1752–1759.

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Dean KR, et al. (2018) Human ectoparasites and the spread of plague in Europe during the Second Pandemic. Proc Natl Acad Sci USA 115:1304–1309

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Park SW, Dushoff J, Earn DJD, Poinar H, Bolker BM (2018) Human ectoparasite transmission of the plague during the Second Pandemic is only weakly supported by proposed mathematical models. Proc Natl Acad Sci USA 115:E7892–E7893.

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Dean, K.R., Krauer, F., Walløe, L., Lingjærde, O.C., Bramanti, B., Stenseth, N.C., Schmid, B.V. REPLY TO PARK ET AL.: Human ectoparasite transmission of plague during the Second Pandemic is still plausible.
(2018) Proceedings of the National Academy of Sciences of the United States of America, 115 (34), pp. E7894-E7895.

(also see Flea News 82 editorial on the controversy on plague).

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Bejcek, J. R., Curtis-Robles, R., Riley, M., Brundage, A., Hamer, S. A., & Hamer, G. L. (2018). Clear Resin Casting of Arthropods of Medical Importance for Use in Educational and Outreach Activities. Journal of Insect Science, 18(2), 34.

Arthropod-related morbidity and mortality represent a major threat to human and animal health. An important component of reducing vector-borne diseases and injuries is training the next generation of medical entomologists and educating the public in proper identification of arthropods of medical importance. One challenge of student training and public outreach is achieving a safe mounting technique that allows observation of morphological characteristics, while minimizing damage to specimens that are often difficult to replace. Although resin-embedded specimens are available from commercial retailers, there is a need for a published protocol that allows entomologists to economically create high-quality resin-embedded arthropods for use in teaching and outreach activities. We developed a detailed protocol using readily obtained equipment and supplies for creating resin-embedded arthropods of many species for use in teaching and outreach activities.

Microbial Symbiotes

Danforth, M., Tucker, J., & Novak, M. (2018). The Deer Mouse (*Peromyscus maniculatus*) as an Enzootic Reservoir of Plague in California. EcoHealth, 1-11.

It has long been theorized that deer mice (*Peromyscus maniculatus*) are a primary reservoir of *Yersinia pestis* in California. However, recent research from other parts of the western USA has implicated deer mice as spillover hosts during epizootic plague transmission. This retrospective study analyzed deer mouse data collected for plague surveillance by public health agencies in California from 1971 to 2016 to help elucidate the role of deer mice in plague transmission. The fleas most commonly found on deer mice were poor vectors of *Y. pestis* and occurred in insufficient numbers to maintain transmission of the pathogen, while fleas whose natural hosts are deer mice were rarely observed and even more rarely found infected with *Y. pestis* on other rodent hosts. Seroprevalence of *Y. pestis* antibodies in deer mice was significantly lower than that of several chipmunk and squirrel species. These analyses suggest that it is unlikely that deer mice play an important role in maintaining plague transmission in California. While they may not be primary reservoirs, results supported the premise that deer mice are occasionally exposed to and infected by *Y. pestis* and instead may be spillover hosts.

Fleas in Art and History

The Tragical History of Dr. Faustus
by Christopher Marlowe
(baptised 26 February 1564, Canterbury, Kent, England – 30 May 1593, Deptford, Kent).

From The Quarto of 1604

"WAGNER. I will teach thee to turn thyself to any thing, to a dog,
or a cat, or a mouse, or a rat, or any thing.

CLOWN. How! a Christian fellow to a dog, or a cat, a mouse,
or a rat! no, no, sir; if you turn me into any thing, let it be
in the likeness of a little pretty frisking flea, that I may be
here and there and every where: O, I'll tickle the pretty wenches'
plackets! I'll be amongst them, i'faith ....

… Enter the SEVEN DEADLY SINS.

Now, Faustus, examine them of their several names and dispositions.

FAUSTUS. What art thou, the first?

PRIDE. I am Pride. I disdain to have any parents. I am like to
Ovid's flea; I can creep into every corner of a wench; sometimes,
like a perriwig, I sit upon her brow; or, like a fan of feathers,
I kiss her lips; indeed, I do—what do I not? But, fie, what a
scent is here! I'll not speak another word, except the ground
were perfumed, and covered with cloth of arras."

[Editor's note: named for Arras, France, "cloth of arras" were Flemish tapestries used as wall hangings, often screening doors and windows.]

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Anonymous. 1887. The Autobiography of a Flea. Edward Avery Publisher, UK.

Born I was—but how, when, or where I cannot say; so I must leave the reader to accept the assertion " per se, " and believe it if he will. One thing is equally certain, the fact of my birth is not one atom less veracious than the reality of these memoirs, and if the intelligent student of, these pages wonders how it came to pass that one in my walk—or perhaps, I should have said jump—in life, became possessed of the learning, observation and power of committing to memory the whole of the wonderful facts and disclosures I am about to relate. I can only remind him that there are intelligences, little suspected by the vulgar, and laws in nature, the very existence of which have not yet been detected by the advanced among the scientific world.

I have heard it somewhere remarked that my province was to get my living by blood sucking. I am not the lowest by any means of that universal fraternity, and if I sustain a precarious existence upon the bodies of those with whom I come in contact, my own experience proves that I do so in a marked and peculiar manner, with a warning of my employment which is seldom given by those in other grades of my profession. But I submit that I have other and nobler aims than the mere sustaining of my being by the contributions of the unwary. I have been conscious of this original defect, and, with a soul far above the vulgar instincts of my race. I jumped by degrees to heights of mental perception and erudition which placed me for ever upon a pinnacle of insect-grandeur.

It is this attainment to learning which I shall evoke in describing the scenes of which I have been a witness—nay, even a partaker. I shall not stop to explain by what means I am possessed of human powers of thinking and observing, but, in my lucubrations, leave you simply to perceive that I possess them and wonder accordingly.

You will thus perceive that I am not common flea; indeed, when it is born in mind the company in which I have been accustomed to mingle, the familiarity with which I have been suffered to treat persons the most exalted, and the opportunities I have possessed to make the most of my acquaintances, the reader will no doubt agree with me that I am in very truth a most wonderful and exalted insect....

... Be that as it may; my task is done, my promise is performed, my memoir is ended, and if it is out of the power of a Flea to point a moral, at least it is not beyond his ability to choose his own pastures.

Having had quite enough of those of which I have discoursed, I did as many are doing, who, although not Fleas, are, nevertheless as I reminded my readers in the commencement of my narrative, Bloodsuckers; —I emigrated.

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On the next page

Giuseppe Maria Crespi (14 March 1665, Bologna, Italy -16 July 1747, Bologna), Woman Looking for Fleas. c. 1720s. Oil on canvas. One of a series, see Flea News 71 for an earlier one.

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Siphonaptera Literature (2018)

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Abdulkareem, B. O., Christy, A. L., & Samuel, U. U. (2019). Prevalence of ectoparasite infestations in owned dogs in Kwara state, Nigeria. Parasite Epidemiology and Control, 4 doi:10.1016/j.parepi.2018.e00079

Abreu-Yanes, E., Martin-Alonso, A., Martin-Carrillo, N., Livia, K. G., Marrero-Gagliardi, A., Valladares, B., . . . Foronda, P. (2018). *Bartonella* in rodents and ectoparasites in the Canary Islands, Spain: New insights into Host–Vector–Pathogen relationships. Microbial Ecology, 75(1), 264-273. doi:10.1007/s00248-017-1022-y

Aguilar Montiel, F., Estrada-Torres, A., Acosta, R., Rubio-Godoy, M., & Vázquez, J. (2018). Host species influence on flea (Siphonaptera) infection parameters of terrestrial micromammals in a temperate forest of Mexico. Parasitology, doi:10.1017/S0031182018001981

Ahn, K. -., Huh, S. -., Seol, S. -., Kim, H. -., Suh, K. -., & Shin, S. (2018). *Ctenocephalides canis* is the dominant flea species of dogs in the Republic of Korea. Parasites and Vectors, 11(1) doi:10.1186/s13071-018-2769-9

Amphlett, A. (2018). Shadow of the third pandemic: A review of the epidemiology, microbiology, ecology, clinical features, molecular diagnosis and treatment of plague. Reviews in Medical Microbiology, 29(3), 110-115. doi:10.1097/MRM.0000000000000141

Andrianaivoarimanana, V., Rajerison, M., & Jambou, R. (2018). Exposure to *Yersinia pestis* increases resistance to plague in black rats and modulates transmission in Madagascar. BMC Research Notes, 11(1) doi:10.1186/s13104-018-3984-3

Anon. Medicines update. (2018). Veterinary Record, 183(8), 249-251. doi:10.1136/vr.k3720

Anon. Oral protection for cats against fleas and ticks. (2018). The Veterinary Record, 183(20), 617.

Anon. Skin and other pathogens: Malaria and plague. (2018). Skin and arthropod vectors (pp. 239-274) doi:10.1016/B978-0-12-811436-0.00007-1

Appelgren, A. S. C., Saladin, V., Richner, H., Doligez, B., & McCoy, K. D. (2018). Gene flow and adaptive potential in a generalist ectoparasite. BMC Evolutionary Biology, 18(1) doi:10.1186/s12862-018-1205-2

Arcà, B., & Ribeiro, J. M. (2018). Saliva of hematophagous insects: a multifaceted toolkit. *Current opinion in insect science* 29: 102-109.

Archer, C. E., Schoeman, M. C., Appleton, C. C., Mukaratirwa, S., Hope, K. J., & Matthews, G. B. (2018). Predictors of *Trypanosoma lewisi* in Rattus norvegicus from Durban, South Africa. Journal of

Parasitology, 104(3), 187-195. doi:10.1645/17-92

Arce S., D.E. Manzoli, M. J. Saravia-Pietropaolo, M. A. Quiroga, L. R. Antoniazzi, M. Lareschi; P. M. Beldomenico. The tropical fowl mite, *Ornithonyssus bursa* (Acari: Macronyssidae): Environmental and host factors associated with its occurrence in Argentine passerine communities. Parasitology Research https://doi.org/10.1007/s00436-018-6025-1

Balasundaram, M. B., Manjunath, M., Baliga, G., & Kapadi, F. (2018). Ocular manifestations of *Rickettsia conorii* in south India. Indian Journal of Ophthalmology, 66(12), 1840-1844. doi:10.4103/ijo.IJO_420_18

Bazanova, L.P., & Nikitin, A. Ya. (2018). Relations between representatives of eight genera of the order Siphonaptera and *Yersinia pestis* from the Tuvinian natural focus of plague. Epidemiology and vaccine prevention, 17 (3 (100). [Базанова, Л. П., & Никитин, А. Я. (2018). Взаимоотношения представителей восьми родов отряда Siphonaptera и *Yersinia pestis* из тувинского природного очага чумы. Эпидемиология и вакцинопрофилактика, 17(3 (100)).]

[*Citellophilus tesquorum altaicus, Frontopsylla elatoides, Neopsylla mana, Oropsylla alaskensis, Rhadinopsylla li transbaicalica* on *Spermophilus undulatus.*]

Beaucournu AD Zewdneh TT Bereket Laudisoit AA & A., 2018. Description of the male of *Ctenophthalmus (Ethioctenophthalmus) vanhoutteae* Beaucournu & Bereket, 2017. Bulletin of the Entomological Society of France, 123: 329-332.

Benelli, G., & Pavela, R. (2018). Beyond mosquitoes—Essential oil toxicity and repellency against bloodsucking insects. Industrial Crops and Products, 117, 382-392. doi:10.1016/j.indcrop.2018.02.072

Beugnet, F., Labuschagne, M., Vos, C. D., Crafford, D., & Fourie, J. (2018). Analysis of *Dipylidium caninum* tapeworms from dogs and cats, or their respective fleas: Part 2. Distinct canine and feline host association with two different *Dipylidium caninum* genotypes. Parasite, 25 doi:10.1051/parasite/2018029

Bland, D. M., Jarrett, C. O., Bosio, C. F., & Hinnebusch, B. J. (2018). Infectious blood source alters early foregut infection and regurgitative transmission of *Yersinia pestis* by rodent fleas. PLoS Pathogens, 14(1) doi:10.1371/journal.ppat.1006859

Boegler, K. A., Atiku, L. A., Enscore, R. E., Apangu, T., Mpanga, J. T., Acayo, S., . . . Eisen, R. J. (2018). Rat fall surveillance coupled with vector control and community education as a plague prevention strategy in the west Nile region, Uganda. American Journal of Tropical Medicine and Hygiene, 98(1), 238-247. doi:10.4269/ajtmh.17-0502

Böhm, A., Meusemann, K., Misof, B., & Pass, G. (2018). Hypothesis on monochromatic vision in scorpionflies questioned by new transcriptomic data. Scientific Reports, 8(1) doi:10.1038/s41598-018-28098-2

Borges, D. A., Moraes, P. D. A., Cardoso, J. D., Oliveira, P. C. D., Yasui, A. M., Fernandes, I. M. P., . . .

Scott, F. B. (2018). Efficacy of a dinotefuran, pyriproxyfen and permethrin combination product against *Ctenocephalides felis felis* (Bouché 1835) (Siphonaptera: Pulicidae) on artificially infested rabbits. Veterinary Parasitology, 259, 74-79. doi:10.1016/j.vetpar.2018.07.006

Borraccini, L., Jorge, M., & Mónica, P. (2018). Association between atopic dermatitis manifestations and positive intradermorreaction in dogs of the city of Rosario, Argentina. [Asociación entre manifestaciones de dermatitis atópica e intradermorreaccion positiva en perros de la ciudad de Rosario, Argentina] Revista Electronica De Veterinaria, 19(5)

Bouchaib, H., Eldin, C., Laroche, M., Raoult, D., & Parola, P. (2018). Tick- and flea-borne rickettsioses in Tizi-ouzou, Algeria: Implications for travel medicine. Travel Medicine and Infectious Disease, 26, 51-57. doi:10.1016/j.tmaid.2018.11.005

Boyes, D. H., & Lewis, O. T. (2019). Ecology of Lepidoptera associated with bird nests in mid-Wales, UK. Ecological Entomology, 44(1), 1-10. doi:10.1111/een.12669

Breitschwerdt, E. B., & Maggi, R. G. (2018). *Bartonella quintana* and *Bartonella vinsonii* subsp. *vinsonii* bloodstream co-infection in a girl from North Carolina, USA. Medical Microbiology and Immunology, doi:10.1007/s00430-018-0563-0

C. Briand, 2017. Collection of Siphonaptera Beaucournu AD: The Siphonaptera of the Armorican. Invertebrates Armoricans, No. 17, pp.1 - 39.

Bron, G. M., Richgels, K. L. D., Samuel, M. D., Poje, J. E., Lorenzsonn, F., Matteson, J. P., . . . Rocke, T. E. (2018). Impact of sylvatic plague vaccine on non-target small rodents in grassland ecosystems. EcoHealth, 15(3), 555-565. doi:10.1007/s10393-018-1334-5

Burazerović, J., Orlova, M., Obradović, M., Ćirović, D., & Tomanović, S. (2018). Patterns of abundance and host specificity of bat ectoparasites in the central Balkans. Journal of Medical Entomology, 55(1), 20-28. doi:10.1093/jme/tjx189

Carithers, D., Dryden, M., Everett, W. R., Gross, S., & Crawford, J. (2018). Assessment of Frontline® Plus efficacy at 24-hour counts against Tampa 2014 isolate *Ctenocephalides felis* flea infestations on cats and dogs on days 1, 7, 14, 21, and 28. International Journal of Applied Research in Veterinary Medicine, 16(1), 52-58.

Carithers, D., Everett, W. R., Gross, S., & Crawford, J. (2018). Comparative efficacy of fipronil/(S)-methoprene/pyriproxyfen (FRONTLINE® gold) and sarolaner (SIMPARICA®) against *Ctenocephalides felis* flea infestations on dogs. International Journal of Applied Research in Veterinary Medicine, 16(1), 27-31.

Cassin Sackett, L. (2018). Does the host matter? Variable influence of host traits on parasitism rates. International Journal for Parasitology, 48(1), 27-39. doi:10.1016/j.ijpara.2017.07.006

Castaño-Vázquez, F., Martínez, J., Merino, S., & Lozano, M. (2018). Experimental manipulation of temperature reduce ectoparasites in nests of blue tits *Cyanistes caeruleus*. Journal of Avian Biology, 49(8) doi:10.1111/jav.01695

Cavalleri, D., Murphy, M., Seewald, W., & Nanchen, S. (2018). A randomized, controlled field study to assess the efficacy and safety of Lotilaner (Credelio™) in controlling fleas in client-owned cats in Europe. Parasites and Vectors, 11(1) doi:10.1186/s13071-018-2971-9

Cavalleri, D., Murphy, M., Seewald, W., & Nanchen, S. (2018). Laboratory evaluation of the efficacy and speed of kill of lotilaner (credelio™) against *Ctenocephalides felis* on cats. Parasites and Vectors, 11(1) doi:10.1186/s13071-018-2972-8

Çelebi, B., Yeşilyurt, M., & Kiliç, S. (2018). A *Rickettsia* case caused by *Rickettsia conorii.* [*Rickettsia conorii*'nin Neden Olduğu Riketsiyal Enfeksiyon Olgusu] Mikrobiyoloji Bulteni, 52(4), 431-438. doi:10.5578/mb.67139

Cevidanes, A., Di Cataldo, S., Vera, F., Lillo, P., & Millán, J. (2018). Molecular detection of vector-borne pathogens in rural dogs and associated *Ctenocephalides felis* fleas (Siphonaptera: Pulicidae) in Easter Island (Chile). Journal of Medical Entomology, 55(6), 1659-1663. doi:10.1093/jme/tjy141

Charrel, R. N., Berenger, J., Laroche, M., Ayhan, N., Bitam, I., Delaunay, P., & Parola, P. (2018). Neglected vector-borne bacterial diseases and arboviruses in the Mediterranean area. New Microbes and New Infections, 26, S31-S36. doi:10.1016/j.nmni.2018.08.015

Chen, Y. -., Huang, C. -., Hsu, J. -., & Wu, W. -. (2018). Development of a larval bioassay method using 96-well microtiter plates for evaluation of susceptibility of the cat fleas (Siphonaptera: Pulicidae) to insecticides. Journal of Medical Entomology, 54(2), 377-381. doi:10.1093/jme/tjw156

Cheng, T., Halper, B., Siebert, J., Cruz-Martinez, L., Chapwanya, A., Kelly, P., . . . Yao, C. (2018). Parasites of small Indian mongoose, *Herpestes auropunctatus*, on St. Kitts, West Indies. Parasitology Research, 117(4), 989-994. doi:10.1007/s00436-018-5773-2

Chochlakis, D., Germanakis, A., Chaliotis, G., Kapetanaki, S., Kalogeraki, L., Gkika, E., . . . Psaroulaki, A. (2018). Potential exposure of humans to *Rickettsia felis* in Greece. Acta Tropica, 178, 40-45. doi:10.1016/j.actatropica.2017.10.020

Clark, N. J., Seddon, J. M., Šlapeta, J., & Wells, K. (2018). Parasite spread at the domestic animal - wildlife interface: Anthropogenic habitat use, phylogeny and body mass drive risk of cat and dog flea (*Ctenocephalides* spp.) infestation in wild mammals. Parasites & Vectors, 11(1), 8. doi:10.1186/s13071-017-2564-z

Cohen, C., Shemesh, M., Garrido, M., Messika, I., Einav, M., Khokhlova, I., . . . Hawlena, H. (2018). Haemoplasmas in wild rodents: Routes of transmission and infection dynamics. Molecular Ecology, 27(18), 3714-3726. doi:10.1111/mec.14826

Contreras, M., Villar, M., Artigas-Jerónimo, S., Kornieieva, L., Mytrofanov, S., & De La Fuente, J. (2018). A reverse vaccinology approach to the identification and characterization of *Ctenocephalides felis* candidate protective antigens for the control of cat flea infestations. Parasites and Vectors, 11(1) doi:10.1186/s13071-018-2618-x

Cooke, B. D., & Mutze, G. J. (2018). How introduced rabbits *Oryctolagus cuniculus* limit the abundance of red kangaroos *Macropus rufus* and other native grazers in Australia. Food Webs, 15 doi:10.1016/j.fooweb.2018.e00079

Coşkun, G., & Çetin, H. (2018). A research about flea (Siphonaptera: Pulicidae) infestation on domestic cats and dogs in winter months, from Antalya, Turkey. Turkiye Parazitolojii Dergisi, 42(4), 277-280. doi:10.5152/tpd.2018.5971

da Silva, G. A. C., Lins, L. A., Irala, M. J. C., Cárcamo, M. C., & Ribeiro, P. B. (2018). EFFECT OF EL NIÑO PHENOMENON ON THE SEASONALITY OF Pulicidae (Siphonaptera) IN DOGS IN A RURAL AREA OF SOUTHERN BRAZIL. *Science And Animal Health*, *6*(1), 72-85.

Dai, R., Wei, B., Xiong, H., Yang, X., Peng, Y., He, J., . . . Li, W. (2018). Human plague associated with Tibetan sheep originates in marmots. PLoS Neglected Tropical Diseases, 12(8) doi:10.1371/journal.pntd.0006635

Danforth, M., Tucker, J., & Novak, M. (2018). The deer mouse (*Peromyscus maniculatus*) as an enzootic reservoir of plague in California. EcoHealth, 1-11. doi:10.1007/s10393-018-1337-2

De Lima, F. C. G., & De Oliveira Porpino, K. (2018). Ectoparasitism and infections in the exoskeletons of large fossil cingulates. PLoS ONE, 13(10) doi:10.1371/journal.pone.0205656

de Sousa, K. C. M., Fernandes, M. P., Herrera, H. M., Freschi, C. R., Machado, R. Z., & André, M. R. (2018). Diversity of piroplasmids among wild and domestic mammals and ectoparasites in Pantanal Wetland, Brazil. Ticks and Tick-Borne Diseases, 9(2), 245-253. doi:10.1016/j.ttbdis.2017.09.010

de Sousa, K. C. M., Herrera, H. M., Rocha, F. L., Costa, F. B., Martins, T. F., Labruna, M. B., . . . André, M. R. (2018). *Rickettsia* spp. among wild mammals and their respective ectoparasites in Pantanal Wetland, Brazil. Ticks and Tick-Borne Diseases, 9(1), 10-17. doi:10.1016/j.ttbdis.2017.10.015

Dean, K. R., Krauer, F., Walløe, L., Lingjærde, O. C., Bramanti, B., Stenseth, N. C., & Schmid, B. V. (2018). Human ectoparasites and the spread of plague in Europe during the second pandemic. Proceedings of the National Academy of Sciences of the United States of America, 115(6), 1304-1309. doi:10.1073/pnas.1715640115

Dean, K. R., Krauer, F., Walløe, L., Lingjærde, O. C., Bramanti, B., Stenseth, N. C., & Schmid, B. V. (2018). REPLY TO PARK ET AL.: Human ectoparasite transmission of plague during the second pandemic is still plausible. Proceedings of the National Academy of Sciences of the United States of America, 115(34), E7894-E7895. doi:10.1073/pnas.1810221115

Dik, B., & Uslu, U. (2018). Ectoparasites of hares (*Lepus europaeus* Pallas) in Konya province, Turkey. Turkish Journal of Veterinary and Animal Sciences, 42(1), 65-72. doi:10.3906/vet-1706-19

Dos Santos Grácio, A. J., & Grácio, M. A. A. (2018). Plague in Madagascar. Tropical Doctor, 48(1), 1-2. doi:10.1177/0049475517749302

Dryden, M. W., Canfield, M. S., Bocon, C., Phan, L., Niedfeldt, E., Kinnon, A., . . . Sun, F. (2018). In-home assessment of either topical fluralaner or topical selamectin for flea control in naturally infested cats in west central Florida, USA. Parasites and Vectors, 11(1) doi:10.1186/s13071-018-2995-1

Duscher, G. G., Hodzić, A., Potkonjak, A., Leschnik, M. W., & Spergser, J. (2018). *Bartonella henselae* and *Rickettsia felis* detected in cat fleas (*Ctenocephalides felis*) derived from eastern Austrian cats. Vector-Borne and Zoonotic Diseases, 18(5), 282-284. doi:10.1089/vbz.2017.2215

Eads, D. A. (2018). Parasitism of Black-Tailed Prairie Dogs by *Linognathoides cynomyis* (Phthiraptera: Polyplacidae). *Journal of medical entomology*, *56*(1), 280-283.

Eads, D. A., Biggins, D. E., Bowser, J., McAllister, J. C., Griebel, R. L., Childers, E., . . . Bly, K. (2018). Resistance to deltamethrin in prairie dog (*Cynomys ludovicianus*) fleas in the field and in the laboratory. Journal of Wildlife Diseases, 54(4), 745-754. doi:10.7589/2017-10-250

Edwards, C. H., Baird, J., Zinser, E., Woods, D. J., Shaw, S., Campbell, E. M., & Bowman, A. S. (2018). RNA interference in the cat flea, *Ctenocephalides felis*: Approaches for sustained gene knockdown and evidence of involvement of dicer-2 and Argonaute2. International Journal for Parasitology, 48(13), 993-1002. doi:10.1016/j.ijpara.2018.04.006

El Hamzaoui, B., Laroche, M., Almeras, L., Bérenger, J. -., Raoult, D., & Parola, P. (2018). Detection of *Bartonella* spp. in fleas by MALDI-TOF MS. PLoS Neglected Tropical Diseases, 12(2) doi:10.1371/journal.pntd.0006189

Fang, H., Liu, L., Zhang, Y., Yang, H., Yan, Y., Ding, X., . . . Yang, R. (2018). BfvR, an AraC-family regulator, controls biofilm formation and pH6 antigen production in opposite ways in *Yersinia pestis* biovar *microtus*. Frontiers in Cellular and Infection Microbiology, 8, 347. doi:10.3389/fcimb.2018.00347

Flatau, R., Segoli, M., Khokhlova, I., & Hawlena, H. (2018). *Wolbachia's* role in mediating its flea's reproductive success differs according to flea origin. FEMS Microbiology Ecology, 94(10) doi:10.1093/femsec/fiy157

Fox, M. W. (2018). Keeping cats indoors. Veterinary Record, 183(8), 267. doi:10.1136/vr.k3689

Fukuto, H. S., Vadyvaloo, V., McPhee, J. B., Poinar, H. N., Holmes, E. C., & Bliska, J. B. (2018). A single amino acid change in the response regulator PhoP, acquired during *Yersinia pestis* evolution, affects PhoP target gene transcription and polymyxin B susceptibility. Journal of Bacteriology, 200(9) doi:10.1128/JB.00050-18

Galloway, T. D. (2018). Biodiversity of Ectoparasites: Lice (Phthiraptera) and Fleas (Siphonaptera). *Insect Biodiversity: Science and Society*, *2*, 457-482.

Galy, A., Loubet, P., Peiffer-Smadja, N., & Yazdanpanah, Y. (2018). The plague: An overview and hot topics. [La peste: mise au point et actualités] Revue De Medecine Interne, 39(11), 863-868. doi:10.1016/j.revmed.2018.03.019

Gelasakis, A. I., Valergakis, G. E., & Arsenos, G. (2018). Health and welfare of indigenous goat breeds from dairy farms in Greece. Sustainable goat production in adverse environments (pp. 223-246) doi:10.1007/978-3-319-71855-2_14

Ghavami, M. B., Haghi, F. P., Alibabaei, Z., Enayati, A. A., & Vatandoost, H. (2018). First report of target site insensitivity to pyrethroids in human flea, *Pulex irritans* (siphonaptera: Pulicidae). Pesticide Biochemistry and Physiology, 146, 97-105. doi:10.1016/j.pestbp.2018.03.004

Ghavami, M. B., Mirzadeh, H., Mohammadi, J., & Fazaeli, A. (2018). Molecular survey of ITS1 spacer and *Rickettsia* infection in human flea, *Pulex irritans*. Parasitology Research, 117(5), 1433-1442. doi:10.1007/s00436-018-5768-z

Ghawar, W., Bettaieb, J., Salem, S., Snoussi, M. -., Jaouadi, K., Yazidi, R., & Ben-Salah, A. (2018). Natural infection of *Ctenodactylus gundi* by *Leishmania major* in Tunisia. Acta Tropica, 177, 89-93. doi:10.1016/j.actatropica.2017.09.022

Goncharov, A.I., & Chursinova, N.V. (2018). About fleas parasitizing on passer sparrows. Russian Ornithological Journal, 27 (1640).

González-Álvarez, V. H., Martínez, A. C., & Adeniran, A. A. (2018). Flea (Siphonaptera: Pulicidae) prevalence and first record of *Ctenocephalides canis* (Curtis, 1826) in domestic dogs in north-central México. *J Dairy Vet Anim Res*, *7*(4), 146-148.

Gopinath, D., Meyer, L., Smith, J., & Armstrong, R. (2018). Topical or oral fluralaner efficacy against flea (*Ctenocephalides felis*) transmission of *Dipylidium caninum* infection to dogs. Parasites and Vectors, 11(1) doi:10.1186/s13071-018-3140-x

Grossmann, N. V., de Sousa, A. S., Cardoso, R. M., & Hoppe, E. G. L. (2018). Parasitological findings and antiparasitic treatment of captive jaguarundis *Herpailurus yagouaroundi* (Carnivora: Felidae) in a conservation center in Brazil. Journal of Threatened Taxa, 10(7), 11916-11919. doi:10.11609/jot.3394.10.7.11916-11919

Grybchuk, D., Kostygov, A. Y., Macedo, D. H., Votýpka, J., Lukeš, J., & Yurchenko, V. (2018). RNA viruses in *Blechomonas* (Trypanosomatidae) and evolution of Leishmaniavirus. MBio, 9(5) doi:10.1128/mBio.01932-18

Guadalupe, M. (2018). Introduction to special issue on canine and feline vector borne diseases and relative parasitic zoonoses: Current status across Europe (ESCCAP). Veterinary Parasitology, 255, 97. doi:10.1016/j.vetpar.2018.04.002

Gupta, R. C., & Anadón, A. (2018). Fipronil. Veterinary toxicology: Basic and clinical principles: Third edition (pp. 533-538) doi:10.1016/B978-0-12-811410-0.00042-8

Guru, P. K., Agarwal, A., & Fritz, A. (2018). A miraculous recovery: *Bartonella henselae* infection following a red ant bite. BMJ Case Reports, 2018 doi:10.1136/bcr-2017-222326

Gutiérrez, R., Cohen, C., Flatau, R., Marcos-Hadad, E., Garrido, M., Halle, S., . . . Harrus, S. (2018). Untangling the knots: Co-infection and diversity of *Bartonella* from wild gerbils and their associated fleas. Molecular Ecology, 27(23), 4787-4807. doi:10.1111/mec.14906

Gutiérrez, R., Markus, B., Carstens Marques de Sousa, K., Marcos-Hadad, E., Mugasimangalam, R. C., Nachum-Biala, Y., . . . Harrus, S. (2018). Prophage-driven genomic structural changes promote *Bartonella* vertical evolution. Genome Biology and Evolution, 10(11), 3089-3103. doi:10.1093/gbe/evy236

Haelewaters, D., Hiller, T., & Dick, C. W. (2018). Bats, bat flies, and fungi: A case of hyperparasitism. Trends in Parasitology, 34(9), 784-799. doi:10.1016/j.pt.2018.06.006

Hamza, L., Mehdi, B., Selma, C., & Idir, B. (2018). Understanding the epidemiology and molecular basis of the insecticides-resistance mechanisms in fleas. *New Microbes and New Infections*.

Harimalala, M., Miarinjara, A., Duchemin, J. -., Ramihangihajason, T., & Boyer, S. (2018). Species diversity and phylogeny of fleas of small terrestrial mammals in the forests of the central highlands of Madagascar. Zootaxa, 4399(2), 181-196. doi:10.11646/zootaxa.4399.2.3

Heglasová, I., Víchová, B., Kraljik, J., Mošanský, L., Miklisová, D., & Stanko, M. (2018). Molecular evidence and diversity of the spotted-fever group *Rickettsia* spp. in small mammals from natural, suburban and urban areas of eastern Slovakia. Ticks and Tick-Borne Diseases, 9(6), 1400-1406. doi:10.1016/j.ttbdis.2018.06.011

Hoogland, J. L., Biggins, D. E., Blackford, N., Eads, D. A., Long, D., Rodriguez, M. R., . . . White, E. M. (2018). Plague in a colony of Gunnison's prairie dogs (*Cynomys gunnisoni*) despite three years of infusions of burrows with 0.05% deltamethrin to kill fleas. Journal of Wildlife Diseases, 54(2), 347-351. doi:10.7589/2017-04-089

Hornok, S., Baneth, G., Grima, A., Takács, N., Kontschán, J., Meli, M. L., . . . Hofmann-Lehmann, R. (2018). Molecular investigations of cat fleas (*Ctenocephalides felis*) provide the first evidence of *Rickettsia felis* in Malta and candidatus *Rickettsia senegalensis* in Israel. New Microbes and New Infections, 25, 3-6. doi:10.1016/j.nmni.2018.05.001

Hornok, S., Beck, R., Farkas, R., Grima, A., Otranto, D., Kontschán, J., . . . Baneth, G. (2018). High mitochondrial sequence divergence in synanthropic flea species (Insecta: Siphonaptera) from Europe and the Mediterranean. Parasites and Vectors, 11(1) doi:10.1186/s13071-018-2798-4

Ivanič Porhajašová, J., Noskovič, J., & Babošová, M. (2018). Biodiversity of epigeic groups in sugar beet stands. [Biodiverzita epigeických skupín v porastoch repy cukrovej] Listy Cukrovarnicke a Reparske, 134(9-10), 310-313.

Jaksche, A. (2018). Safe in the countryside: Ticks and fleas can be dangerous for two- and four-legged animals. [Ungefährdet im Grünen: Zecken und Flöhe können für Zwei- und Vierbeiner gefährlich warden] Deutsche Apotheker Zeitung, 158(37)

Kamani, J., Harrus, S., Nachum-Biala, Y., Salant, H., Mumcuoglu, K. Y., & Baneth, G. (2018). Pathogenic and endosymbiont apicomplexans in *Ctenocephalides felis* (Siphonaptera: Pulicidae) from cats in Jerusalem, Israel. Comparative Immunology, Microbiology and Infectious Diseases, 57, 29-33. doi:10.1016/j.cimid.2018.03.002

Kandi, V. (2018). Tungiasis Presenting as Onychomycosis: Probably the First Report of Flea Infestation of the Nail Observed Using Modified Potassium Hydroxide Mount Technique. *Cureus*, *10*(3).

Kapia, S. (2018). Detection of *Leptospira* and seasonal prevalence of fleas collected from small mammals in Mukwe constituency, Kavango East region of Namibia (Degree of Master of Science (Biology), University of Namibia).

Karchesy, J. J., Kelsey, R. G., & González-Hernández, M. P. (2018). Yellow-cedar, *Callitropsis (Chamaecyparis) nootkatensis*, secondary metabolites, biological activities, and chemical ecology.

Journal of Chemical Ecology, 44(5), 510-524. doi:10.1007/s10886-018-0956-y

Keskin, A., Hastriter, M. W., & Beaucournu, J. (2018). Fleas (Siphonaptera) of Turkey: Species composition, geographical distribution and host associations. Zootaxa, 4420(2), 211-228. doi:10.11646/zootaxa.4420.2.4

Khalip, J. (2018). The last animal at the end of the world. European Romantic Review, 29(3), 321-332. doi:10.1080/10509585.2018.1465706

Khelfaoui, F., Kebaci, A., & Benyacoub, S. (2018). New data on Insecta and Acarina parasitizing bats (Mammalia: Chiroptera) in Numidia, eastern Algeria. Bulletin De La Societe Zoologique De France, 143(2), 63-73.

Kho, K. L., & Tay, S. T. (2018). Identification of *Rickettsia*l Infections (*Rickettsia* sp. TH2014) in *Ctenocephalides orientis* Fleas (Siphonaptera: Pulicidae). *Journal of medical entomology.*

Khosravani, M. (2018). The fauna and perspective of Rodentia ectoparasites in Iran relying on their roles within public health and veterinary characteristics. Journal of Parasitic Diseases, 42(1) doi:10.1007/s12639-017-0959-6

Kotti, B. K., & Kot, L. A. (2018). Patterns of formation of the fauna of fleas (Siphonaptera) parasitizing mammals in the Stavropol Upland and adjacent territories. Entomological Review, 98(1), 117-124. doi:10.1134/S0013873818010104

Krasnov, B. R., Shenbrot, G. I., van der Mescht, L., Warburton, E. M., & Khokhlova, I. S. (2018). The latitudinal, but not the longitudinal, geographic range positions of haematophagous ectoparasites demonstrate historical signatures. International Journal for Parasitology, 48(9-10), 743-749. doi:10.1016/j.ijpara.2018.03.004

Krasnov, B. R., Shenbrot, G. I., Van Der Mescht, L., Warburton, E. M., & Khokhlova, I. S. (2018). Phylogenetic heritability of geographic range size in haematophagous ectoparasites: Time of divergence and variation among continents. Parasitology, 145(12), 1623-1632. doi:10.1017/S0031182018000550

Krasnov, B. R., Shenbrot, G. I., Warburton, E. M., Van Der Mescht, L., Surkova, E. N., Medvedev, S. G., . . . Khokhlova, I. S. (2018). Species and site contributions to β -diversity in fleas parasitic on the palearctic small mammals: Ecology, geography and host species composition matter the most. Parasitology, , 1-9. doi:10.1017/S0031182018001944

Krasnov, B. R., Shenbrot, G. S., van der Mescht, L., Warburton, E. M., & Khokhlova, I. S. (2019). Phylogenetic and compositional diversity are governed by different rules: A study of fleas parasitic on small mammals in four biogeographic realms. Ecography, doi:10.1111/ecog.04224

Kruse, N. B. (2018). The jumping flea diaspora: An ethnographic exploration of music and culture in a hawaiian ‘ukulele club. Music Education Research, 20(5), 603-615. doi:10.1080/14613808.2018.1516745

Kuehn, B. M. (2018). News from the centers for disease control and prevention. JAMA - Journal of the American Medical Association, 319(24), 2471. doi:10.1001/jama.2018.7441

Kuntz, E. A., & Kammanadiminti, S. (2018). Safety of Lotilaner flavoured chewable tablets (Credelio™) after oral administration in cats. Parasites and Vectors, 11(1) doi:10.1186/s13071-018-2969-3

Kwak, M. L. (2018). Australia's vanishing fleas (Insecta: Siphonaptera): A case study in methods for the assessment and conservation of threatened flea species. Journal of Insect Conservation, 22(3-4), 545-550. doi:10.1007/s10841-018-0083-7

Labuschagne, M., Beugnet, F., Rehbein, S., Guillot, J., Fourie, J., & Crafford, D. (2018). Analysis of *Dipylidium caninum* tapeworms from dogs and cats, or their respective fleas: Part 1. molecular characterization of *Dipylidium caninum*: Genetic analysis supporting two distinct species adapted to dogs and cats. Parasite, 25 doi:10.1051/parasite/2018028

Lappin, M. R. (2018). Update on flea and tick associated diseases of cats. Veterinary Parasitology, 254, 26-29. doi:10.1016/j.vetpar.2018.02.022

Lareschi, M., Venzal, J. M., Nava, S., Mangold, A. J., Portillo, A., Palomar-Urbina, A. M., & Oteo-Revuelta, J. A. (2018). The human flea *Pulex irritans* (Siphonaptera: Pulicidae) in northwestern Argentina, with an investigation of *Bartonella* and *Rickettsia* spp. Revista Mexicana De Biodiversidad, 89(2), 375-381. doi:10.22201/ib.20078706e.2018.2.2392

Lareschi M. 2018.Description of the males of *Androlaelaps misionalis* and *Androlaelaps ulysespardinasi* (Acari: Parasitiformes: Laelapidae) parasitic of sigmondontine rodents from northeastern Argentina. Journal of Parasitology 104 (4): 372-376. https://doi.org/10.1645/18-1

Lareschi M., JM Venzal, S Nava, AJ Mangold, A Portillo, AM Palomar Urbina & JA Oteo Revuelta. 2018. The human flea *Pulex irritans* Linnaeus, 1758 (Siphonaptera: Pulicidae) and an investigation of *Bartonella* and *Rickettsia* in northwestern Argentina. Revista Mexicana de Biodiversidad89: 375-381

Larsen, K. S., Sciuto, M., & Dahl, J. (2018). The use of phoxim and bendiocarb for control of fleas in farmed mink (*Mustela vison*). Acta Veterinaria Scandinavica, 60(1) doi:10.1186/s13028-018-0412-6

Lavan, R., Armstrong, R., Burgio, F., & Tunceli, K. (2018). Duration of annual canine flea and tick protection provided by dog owners in Spain. Parasites and Vectors, 11(1) doi:10.1186/s13071-018-3043-x

Lavan, R., Armstrong, R., Tunceli, K., & Normile, D. (2018). Dog owner flea/tick medication purchases in the USA. Parasites and Vectors, 11(1) doi:10.1186/s13071-018-3142-8

Lavan, R., Heaney, K., Vaduvoor, S. R., & Tunceli, K. (2018). A comparative analysis of heartworm medication use patterns for dogs that also receive ectoparasiticides. Parasites and Vectors, 11(1) doi:10.1186/s13071-018-3076-1

Lebon, W., Franc, M., Bouhsira, E., Lienard, E., Murray, M., Carithers, D., & Beugnet, F. (2018). Prevention of flea egg development in a simulated home environment by Frontline® Gold (fipronil, (S)-methoprene, pyriproxyfen) applied topically to cats. International Journal of Applied Research in Veterinary Medicine, 16(1), 74-80.

Liljesthrom G, Lareschi M. Predicting species richness of ectoparasites of wild rodents from the Río de la Plata coastal wetlands, Argentina. Parasitology Research 117: 2507–2520 https://doi.org/10.1007/s00436-018-5940-5

Liu, Y. (2018). Sowing dragon's teeth and harvesting fleas? - A historical inquiry into China's education-aid programs since the 1970s. African and Asian Studies, 17(4), 394-415. doi:10.1163/15692108-12341408

Lonsdale, O., & Locke, M. M. (2018). Name-bearing type specimens in the Canadian National Collection of insects, arachnids & nematodes (CNC): Blattodea, Dermaptera, Notoptera, Mecoptera, Megaloptera, Myriapoda, Neuroptera, Odonata, Orthoptera, Phthiraptera, Pseudoscorpiones, Psocoptera, Raphidioptera & Siphonaptera. Zootaxa, 4526(2), 101-126. doi:10.11646/zootaxa.4526.2.1

López-Berrizbeitia, M. F., Sanchez, J., Barquez, R. M., & Díaz, M. M. (2018). Descriptions of two new species of flea of the genus *Plocopsylla* in northwestern Argentina. Medical and Veterinary Entomology, 32(3), 334-345. doi:10.1111/mve.12303

López-Pérez, A. M., Gage, K., Rubio, A. V., Montenieri, J., Orozco, L., & Suzan, G. (2018). Drivers of flea (Siphonaptera) community structure in sympatric wild carnivores in northwestern Mexico. Journal of Vector Ecology, 43(1), 15-25. doi:10.1111/jvec.12278

Loyola, S., Flores-Mendoza, C., Torre, A., Kocher, C., Melendrez, M., Luce-Fedrow, A., . . . Leguia, M. (2018). *Rickettsia asembonensis* characterization by multilocus sequence typing of complete genes, Peru. Emerging Infectious Diseases, 24(5), 931-933. doi:10.3201/eid2405.170323

Ma, Y., Li, H., He, J., Zhao, Y., Yang, H., Lu, L., & Liu, Q. (2018). Molecular identification and phylogenetic analysis of 44 species of fleas in Qinghai Province, western China based on mtDNA COI gene sequence. *Acta Entomologica Sinica*, *61*(4), 488-497.

Maaz, D., Krücken, J., Blümke, J., Richter, D., McKay-Demeler, J., Matuschka, F. -., . . . Von Samson-Himmelstjerna, G. (2018). Factors associated with diversity, quantity and zoonotic potential of ectoparasites on urban mice and voles. PLoS ONE, 13(6) doi:10.1371/journal.pone.0199385

Machado, I. B., Bitencourth, K., Cardoso, K. M., Oliveira, S. V., Santalucia, M., Marques, S. F. F., . . . GazêTa, G. S. (2018). Diversity of Rickettsiae and potential vectors of spotted fever in an area of epidemiological interest in the Cerrado biome, midwestern Brazil. Medical and Veterinary Entomology, 32(4), 481-489. doi:10.1111/mve.12315

Maronde, L., Losdat, S., & Richner, H. (2018). Do parasites and antioxidant availability affect begging behaviour, growth rate and resistance to oxidative stress? Journal of Evolutionary Biology, 31(6), 904-913. doi:10.1111/jeb.13274

Martino, P. E., Radman, N. E., Gamboa, M. I., Samartino, L. E., & Parrado, E. J. (2018). Ectoparasites from some *Myocastor coypus (*Molina, 1782) populations (coypus or nutria) in Argentina. [Ectoparasitas de algumas populações de *Myocastor coypus* (Molina, 1782) (Coypus ou Nutria) na Argentina] Revista Brasileira De Parasitologia Veterinaria, 27(2), 254-257. doi:10.1590/S1984-296120180026

Matsuura, H., & Yokota, K. (2018). Case report: Family cluster of Japanese spotted fever. American

Journal of Tropical Medicine and Hygiene, 98(3), 835-837. doi:10.4269/ajtmh.17-0199

Mazurek, Ł., Winiarczyk, S., & Adaszek, Ł. (2018). Cat scratch disease as zoonosis: Pathogenesis, clinical symptoms, diagnosis. [Choroba kociego pazura jako zoonoza: Patogeneza, objawy kliniczne, rozpoznawanie] Medycyna Weterynaryjna, 74(11), 693-696. doi:10.21521/mw.5997

McAllister, C. T., Durden, L. A., Robison, H. W., & Connior, M. B. (2017). The Fleas (Arthropoda: Insecta: Siphonaptera) of Arkansas. Journal of the Arkansas Academy of Science, 71(1), 69-76.

McKee, C. D., Osikowicz, L. M., Schwedhelm, T. R., Maes, S. E., Enscore, R. E., Gage, K. L., & Kosoy, M. Y. (2018). Acquisition of *Bartonella elizabethae* by Experimentally Exposed Oriental Rat Fleas (*Xenopsylla cheopis*; Siphonaptera, Pulicidae) and Excretion of *Bartonella* DNA in Flea Feces. *Journal of medical entomology.*

Medvedev, S. G. (2018). Morphological diversity of fleas' structures (Siphonaptera). part 5: Features of thoracic combs and legs. Entomological Review, 98(7), 819-825. doi:10.1134/S0013873818070035

Medvedev, S. G. (2018). Morphological diversity of the skeletal structures and problems of classification of fleas (Siphonaptera). part 6. Entomological Review, 98(1), 10-20. doi:10.1134/S0013873818010037

Melman, S. D., Ettestad, P. E., VinHatton, E. S., Ragsdale, J. M., Takacs, N., Onischuk, L. M., . . . Petersen, J. M. (2018). Human case of bubonic plague resulting from the bite of a wild Gunnison's prairie dog during translocation from a plague-endemic area. Zoonoses and Public Health, 65(1), e254-e258. doi:10.1111/zph.12419

Merola, V. M., & Eubig, P. A. (2018). Toxicology of avermectins and milbemycins (macrocyclic lactones) and the role of P-glycoprotein in dogs and cats. Veterinary Clinics of North America - Small Animal Practice, 48(6), 991-1012. doi:10.1016/j.cvsm.2018.07.002

Millán, J., Travaini, A., Cevidanes, A., Sacristán, I., & Rodríguez, A. (2019). Assessing the natural circulation of canine vector-borne pathogens in foxes, ticks and fleas in protected areas of Argentine Patagonia with negligible dog participation. International Journal for Parasitology: Parasites and Wildlife, 8, 63-70. doi:10.1016/j.ijppaw.2018.11.007

Mongkol, N., Suputtamongkol, Y., Taweethavonsawat, P., & Foongladda, S. (2018). Molecular evidence of *Rickettsia* in human and dog blood in Bangkok. Vector-Borne and Zoonotic Diseases, 18(6), 297-302. doi:10.1089/vbz.2017.2180

Montoya, A., García, M., Gálvez, R., Checa, R., Marino, V., Sarquis, J., . . . Miró, G. (2018). Implications of zoonotic and vector-borne parasites to free-roaming cats in central Spain. Veterinary Parasitology, 251, 125-130. doi:10.1016/j.vetpar.2018.01.009

Moskvina, T. V., Izrailskaia Kharitonova, A. V., & Tsybulsky, A. V. (2018). Parasites of stray and client-owned domestic cats in urban areas in Russia during 2000-2015 years. Tropical Biomedicine, 35(1), 267-279.

Mueller, M. K., Chubb, S., Wolfus, G., & McCobb, E. (2018). Assessment of canine health and preventative care outcomes of a community medicine program. Preventive Veterinary Medicine, 157, 44-49. doi:10.1016/j.prevetmed.2018.05.016

Mujeeb-ur-Rahman, M., Baloch, J. A., Arijo, A. G., Kachiwal, A. B., & Nida, P. (2018). Studies on the prevalence of ectoparasites in owned dogs and major risk infestation to human health in Karachi, Sindh Pakistan. *Proceedings of Parasitology*, (65), 19-29.

Müller, A., Rodríguez, E., Walker, R., Bittencourt, P., Pérez-Macchi, S., Gonçalves, L. R., . . . André, M. R. (2018). Occurrence and genetic diversity of *Bartonella* spp. (Rhizobiales: Bartonellaceae) and *Rickettsia* spp. (Rickettsiales: Rickettsiaceae) in cat fleas (Siphonaptera: Pulicidae) from Chile. Journal of Medical Entomology, 55(6), 1627-1632. doi:10.1093/jme/tjy124

Mullins, K. E., Maina, A. N., Krueger, L., Jiang, J., Cummings, R., Drusys, A., . . . Richards, A. L. (2018). Rickettsial infections among cats and cat fleas in Riverside County, California. American Journal of Tropical Medicine and Hygiene, 99(2), 291-296. doi:10.4269/ajtmh.17-0706

Munkhzul, T., Murdoch, J. D., & Reading, R. P. (2018). Ectoparasites on meso-carnivores in the desert-steppe of Mongolia. *Mongolian Journal of Biological Sciences*, *16*(1), 43-48.

Mutebi, F., Krücken, J., von Samson-Himmelstjerna, G., Waiswa, C., Mencke, N., Eneku, W., . . . Feldmeier, H. (2018). Animal and human tungiasis-related knowledge and treatment practices among animal keeping households in Bugiri District, south-eastern Uganda. Acta Tropica, 177, 81-88. doi:10.1016/j.actatropica.2017.10.003

Nadein, K., & Betz, O. (2018). Jumping mechanisms and performance in beetles. II. Weevils (Coleoptera: Curculionidae: Rhamphini). *Arthropod structure & development*, *47*(2), 131-143.

Nakasu, A., Ishimine, T., Yasumoto, H., Tengan, T., & Mototake, H. (2018). Infective endocarditis associated with *Bartonella henselae*: A case series. IDCases, 12, 127-129. doi:10.1016/j.idcr.2018.04.011

Narladkar, B. W. (2018). Projected economic losses due to vector and vector-borne parasitic diseases in livestock of India and its significance in implementing the concept of integrated practices for vector management. Veterinary World, 11(2), 151-160. doi:10.14202/vetworld.2018.151-160

Nelson, C. A., Moore, A. R., Perea, A. E., & Mead, P. S. (2018). Cat scratch disease: U.S. clinicians' experience and knowledge. Zoonoses and Public Health, 65(1), 67-73. doi:10.1111/zph.12368

Nelson, K., Maina, A. N., Brisco, A., Foo, C., Croker, C., Ngo, V., . . . Wekesa, J. W. (2018). A 2015 outbreak of flea-borne rickettsiosis in San Gabriel Valley, Los Angeles County, California. PLoS Neglected Tropical Diseases, 12(4) doi:10.1371/journal.pntd.0006385

Nguyen, V. K., Parra-Rojas, C., & Hernandez-Vargas, E. A. (2018). The 2017 plague outbreak in Madagascar: Data descriptions and epidemic modelling. Epidemics, 25, 20-25. doi:10.1016/j.epidem.2018.05.001

Noden, B. H., Martin, J., Carrillo, Y., Talley, J. L., & Ochoa-Corona, F. M. (2018). Development of a loop-mediated isothermal amplification (LAMP) assay for rapid screening of ticks and fleas for spotted fever group *Rickettsia*. PLoS ONE, 13(2) doi:10.1371/journal.pone.0192331

Nyirenda, S. S., Hang'ombe, B. M., Simulundu, E., Mulenga, E., Moonga, L., Machang'u, R. S., . . . Kilonzo, B. S. (2018). Molecular epidemiological investigations of plague in eastern province of Zambia. BMC Microbiology, 18(1) doi:10.1186/s12866-017-1146-8

Nziza, J., Tumushime, J. C., Cranfield, M., Ntwari, A. E., Modrý, D., Mudakikwa, A., . . . Šlapeta, J. (2018). Fleas from domestic dogs and rodents in Rwanda carry *Rickettsia asembonensis* and *Bartonella tribocorum*. Medical and Veterinary Entomology, doi:10.1111/mve.12340

Ortega Casanueva, C. (2018). Insect bite allergy. [Alergia a la picadura de insectos] Pediatria Integral, 22(3), 138-146.

Orlova, M. V., & Orlov, O. L. (2018). Contribution to the ectoparasite fauna of bats (Chiroptera: Vespertilionidae, Rhinolophidae) of Crimea. *Entomological Review*, *98*(3), 319-323.

Ortiz, P. A., Garcia, H. A., Lima, L., da Silva, F. M., Campaner, M., Pereira, C. L., . . . Teixeira, M. M. G. (2018). Diagnosis and genetic analysis of the worldwide distributed Rattus-borne *Trypanosoma (Herpetosoma) lewisi* and its allied species in blood and fleas of rodents. Infection, Genetics and Evolution, 63, 380-390. doi:10.1016/j.meegid.2017.09.001

Otranto, D. (2018). Arthropod-borne pathogens of dogs and cats: From pathways and times of transmission to disease control. Veterinary Parasitology, 251, 68-77. doi:10.1016/j.vetpar.2017.12.021

Panagiotakopulu, E., Buckland, P. C., & Wickler, S. (2018). Is there anybody in there? Entomological evidence from a boat burial at Øksnes in Vesterålen, northern Norway. PLoS ONE, 13(7) doi:10.1371/journal.pone.0200545

Peres, G., Yugar, L. B. T., & Haddad Junior, V. (2018). Breakfast, lunch, and dinner sign: A hallmark of flea and bedbug bites. Anais Brasileiros De Dermatologia, 93(5), 759-760. doi:10.1590/abd1806-4841.20187384

Perez, C., Hamouda, D., & Karnath, B. (2018). Murine typhus: A life-threatening presentation of a case in Galveston, Texas. The American Journal of Case Reports, 19, 1503-1506. doi:10.12659/AJCR.909369

Phoosangwalthong, P., Hii, S. F., Kamyingkird, K., Kengradomkij, C., Pinyopanuwat, N., Chimnoi, W., . . . Inpankaew, T. (2018). Cats as potential mammalian reservoirs for *Rickettsia* sp. genotype RF2125 in Bangkok, Thailand. Veterinary Parasitology: Regional Studies and Reports, 13, 188-192. doi:10.1016/j.vprsr.2018.07.001

Pielowska, A., Sontag, E., & Szadziewski, R. (2018). Haematophagous arthropods in Baltic amber. Annales Zoologici, 68(2), 237-249. doi:10.3161/00034541ANZ2018.68.2.003

Poché, D. M., Torres-Poché, Z., Yeszhanov, A., Poché, R. M., Belyaev, A., Dvořák, V., . . . Aimakhanov, B. (2018). Field evaluation of a 0.005% fipronil bait, orally administered to *Rhombomys opimus*, for control of fleas (Siphonaptera: Pulicidae) and phlebotomine sand flies (Diptera: Psychodidae) in the central Asian Republic of Kazakhstan. PLoS Neglected Tropical Diseases, 12(7) doi:10.1371/journal.pntd.0006630

Portillo, A., Ruiz-Arrondo, I., & Oteo, J. A. (2018). Arthropods as vectors of transmisible diseases in

Spain. [Artrópodos vectores en España y sus enfermedades transmisibles] Medicina Clinica, 151(11), 450-459. doi:10.1016/j.medcli.2018.06.021

Pramestuti, N., Umniyati, S. R., Mulyaningsih, B., Widiastuti, D., & Raharjo, J. (2018). Evidence of *Rickettsia typhi* in rat fleas of various habitat and the potential transmission of murine typhus in Banjarnegara, Central Java, Indonesia. Indian Journal of Public Health Research and Development, 9(8), 1548-1554. doi:10.5958/0976-5506.2018.00952.X

Prospero Ponce, C. M., Malik, A. I., Vickers, A., Chevez-Barrios, P., & Lee, A. G. (2018). Orbital inflammatory syndrome secondary to flea bite. Ophthalmic Plastic and Reconstructive Surgery, 34(4), e115-e118. doi:10.1097/IOP.0000000000001115

Radzijevskaja, J., Kaminskiene, E., Lipatova, I., Mardosaite-Busaitiene, D., Balčiauskas, L., Stanko, M., & Paulauskas, A. (2018). Prevalence and diversity of *Rickettsia* species in ectoparasites collected from small rodents in Lithuania. Parasites and Vectors, 11(1) doi:10.1186/s13071-018-2947-9

Rahelinirina, S., Harimalala, M., Margueron, T., Ramihangihajason, T., Mansotte, F., Rajerison, M., . . . Boyer, S. (2018). Risk of maritime introduction of plague from Madagascar to Mayotte. Acta Tropica, 187, 140-143. doi:10.1016/j.actatropica.2018.07.029

Rajonhson, D. M., Miarinjara, A., Rahelinirina, S., Rajerison, M., Boyer, S., & Eisen, R. (2018). Effectiveness of fipronil as a systemic control agent against *Xenopsylla cheopis* (Siphonaptera: Pulicidae) in Madagascar. Journal of Medical Entomology, 54(2), 411-417. doi:10.1093/jme/tjw200

Ranjan, S., Young, D., & Sun, F. (2018). A single topical fluralaner application to cats and to dogs controls fleas for 12 weeks in a simulated home environment. Parasites and Vectors, 11(1) doi:10.1186/s13071-018-2927-0

Rasnitsyn, A. P., & Strelnikova, O. D. (2018). Digestive system of the early Cretaceous insect *Saurophthirus longipes* Ponomarenko (Insecta, ?Aphaniptera, Aaurophthiroidea). Paleontological Journal, 52(2), 146-154. doi:10.1134/S0031030118020119

Rauch, J., Eisermann, P., Noack, B., Mehlhoop, U., Muntau, B., Schäfer, J., & Tappe, D. (2018). Typhus group Rickettsiosis, Germany, 2010–2017. Emerging Infectious Diseases, 24(7), 1213-1220. doi:10.3201/eid2407.180093

Rauch, J., Sothmann, P., Aldrich, C., Hogan, B., Owusu-Dabo, E., May, J., . . . Tappe, D. (2018). Serum cytokine responses in *Rickettsia felis* infected febrile children, Ghana. Medical Microbiology and Immunology, 207(3-4), 243-248. doi:10.1007/s00430-018-0544-3

Razali, N. B., Shamsudin, N., Rahaniza, A. M. J., Yaakop, S., Khoo, J. J., & Mohd-Taib, F. S. (2018). Ectoparasites (ticks and mites) prevalence on small to medium-sized mammals associated with habitat condition in Kemasul, Pahang. Serangga, 23(1), 72-88.

Rennoll, S. A., Rennoll-Bankert, K. E., Guillotte, M. L., Lehman, S. S., Driscoll, T. P., Beier-Sexton, M., . . . Azad, A. F. (2018). The cat flea (*Ctenocephalides felis*) immune deficiency signaling pathway regulates *Rickettsia typhi* infection. Infection and Immunity, 86(1) doi:10.1128/IAI.00562-17

Rodríguez-Cerdeira, C., Gregorio, M. C., & Guzman, R. A. (2018). *Dermatobia hominis* infestation misdiagnosed as abscesses in a traveler to Spain. Acta Dermatovenerologica Croatica, 26(3), 267-269.

Rodríguez-Pastor, R., Escudero, R., Lambin, X., Vidal, M. D., Gil, H., Jado, I., . . . Mougeot, F. (2018). Zoonotic pathogens in fluctuating common vole (*Microtus arvalis*) populations: Occurrence and dynamics. Parasitology, doi:10.1017/S0031182018001543

Rodríguez-Zapata, M., & Martínez, L. S. (2018). Infecciones bacterianas con lesiones cutáneas y fiebre. infecciones por *Rickettsia*. Medicine (Spain), 12(59), 3465-3473. doi:10.1016/j.med.2018.06.012

Rohdich, N., Zschiesche, E., Wolf, O., Loehlein, W., Kirkova, Z., Iliev, P., . . . Roepke, R. K. A. (2018). A randomized, blinded, controlled, multi-centered field study assessing the treatment of gastrointestinal nematode infections in cats with fluralaner plus moxidectin spot-on solution (Bravecto® Plus). Parasites & Vectors, 11(1), 589. doi:10.1186/s13071-018-3169-x

Rohdich, N., Zschiesche, E., Wolf, O., Loehlein, W., Pobel, T., Gil, M. J., & Roepke, R. K. A. (2018). Field effectiveness and safety of fluralaner plus moxidectin (Bravecto® Plus) against ticks and fleas: A European randomized, blinded, multicenter field study in naturally-infested client-owned cats. Parasites & Vectors, 11(1), 598. doi:10.1186/s13071-018-3175-z

Roth, T., Lane, R. S., Foley, J., & Eisen, R. (2018). A molecular survey for *Francisella tularensis* and *Rickettsia* spp. in *Haemaphysalis leporispalustris* (Acari: Ixodidae) in northern California. Journal of Medical Entomology, 54(2), 492-495. doi:10.1093/jme/tjw202

Roth, T., Sammak, R., & Foley, J. (2018). Prevalence and seasonality of fleas associated with California ground squirrels and the potential risk of tularemia in an outdoor non-human primate research facility. Journal of Medical Entomology, 55(2), 452-458. doi:10.1093/jme/tjx201

Rothenburger, J. L., Himsworth, C. G., Nemeth, N. M., Pearl, D. L., & Jardine, C. M. (2018). Beyond abundance: How microenvironmental features and weather influence *Bartonella tribocorum* infection in wild *Norway* rats (*Rattus norvegicus*). Zoonoses and Public Health, 65(3), 339-351. doi:10.1111/zph.12440

Russell, R. E., Abbott, R. C., Tripp, D. W., & Rocke, T. E. (2018). Local factors associated with on-host flea distributions on prairie dog colonies. Ecology and Evolution, 8(17), 8951-8972. doi:10.1002/ece3.4390

Rust, M. K., Hemsarth, W. L. H., & Eisen, R. (2018). Intrinsic activity of IGRs against larval cat fleas. Journal of Medical Entomology, 54(2), 418-421. doi:10.1093/jme/tjw201

Rust, M. K., & Hemsarth, W. L. H. (2019). Synergism of Adulticides and Insect Growth Regulators Against Larval Cat Fleas (Siphonaptera: Pulicidae). Journal of Medical Entomology.

Sanchez, J. P., Ezquiaga, M. C., & Ruiz, M. (2018). Fleas (Insecta: Siphonaptera) with public health relevance in domestic pigs (Artiodactyla: Suidae) from Argentina. Zootaxa, 4374(1), 144-150. doi:10.11646/zootaxa.4374.1.8

Sanchez, J. P., Travaini, A., Rodríguez, A., & Lareschi, M. (2018). Primer registro de pulgas (Insecta, Siphonaptera) en poblaciones de *Lycalopex* (Carnivora, Canidae) de la Patagonia

Argentina. *Mastozool. Neotrop*, *25*, 251-255.

Sanchez, J., & Lareschi, M. (2018). Diversity, distribution and parasitism rates of fleas (Insecta: Siphonaptera) on sigmodontine rodents (Cricetidae) from Argentinian Patagonia. Bulletin of Entomological Research, , 1-12. doi:10.1017/S0007485318000196

Sanchez, J., Lareschi, M., Salazar-Bravo, J., & Gardner, S. L. (2018). Fleas of the genus *Neotyphloceras* associated with rodents from Bolivia: New host and distributional records, description of a new species and remarks on the morphology of *Neotyphloceras rosenbergi*. Medical and Veterinary Entomology, 32(4), 462-472. doi:10.1111/mve.12314

Sanchez J.P. & Y. Gurovich. 2018. Fleas (Insecta: Siphonaptera) Associated to the endangered Neotropical marsupial Monito del Monte (*Dromiciops gliroides* Microbiotheria: Microbiotheriidae) in its Southernmost Population of Argentina. Mastozoología Neotropical 25(1): 257-262.

Santos, N. J., Delabie, J. H. C., Saraiva, D. G., Nieri-Bastos, F. A., Martins, T. F., Labruna, M. B., . . . Franke, C. R. (2018). Effect of forest cover on ectoparasites of small mammals in the Brazilian Atlantic forest. Oecologia Australis, 22(3), 279-290. doi:10.4257/oeco.2018.2203.06

Sarikloglou, E., Goula, A., Sidiropoulos, C., Tsolia, M., & Papa, A. (2018). Murine typhus with marked thrombocytopenia in a child in northern Greece and literature review. Japanese Journal of Infectious Diseases, 71(5), 368-369. doi:10.7883/yoken.JJID.2018.091

Sariyeva, G., Abdel, Z., Shabunin, A., Sagiyev, Z., Abdikarimov, S., Bazarkanova, G., . . . Almazbek, D. (2018). Current status of the Sari-Dzhas natural focus of plague, Kyrgyzstan: Epizootic activity and marmot population. Vector-Borne and Zoonotic Diseases, 18(10), 524-532. doi:10.1089/vbz.2017.2200

Schachterle, J. K., Stewart, R. M., Schachterle, M. B., Calder, J. T., Kang, H., Prince, J. T., & Erickson, D. L. (2018). *Yersinia pseudotuberculosis* BarA-UvrY two-component regulatory system represses biofilms via CsrB. Frontiers in Cellular and Infection Microbiology, 8(SEP) doi:10.3389/fcimb.2018.00323

Scharf, W. C. (2018). New Michigan Tick (Acari: Ixodidae) and Flea (Siphonaptera: Ceratophyllidae) Records From Colonial Nesting Birds. *The Great Lakes Entomologist*, *33*(2), 8.

Scherber, C., Andert, H., Niedringhaus, R., & Tscharntke, T. (2018). A barrier island perspective on species–area relationships. *Ecology and Evolution*.

Seckerdieck, F., & Mueller, R. S. (2018). Recurrent pyoderma and its underlying primary diseases: A retrospective evaluation of 157 dogs. Veterinary Record, 182(15), 434. doi:10.1136/vr.104420

Seyyed-Zadeh, S. J., Bozorg-Omid, F., Telmadarraiy, Z., Terenius, O., & Chavshin, A. R. (2018). Evidence for the presence of *Ctenocephalides orientis* in livestock dwellings in northwest Iran. Medical and Veterinary Entomology, 32(3), 383-387. doi:10.1111/mve.12308

Shabir, M., Yatoo, M. I., Kanwar, M. S., Kubrevi, S. S., Dar, R., & Angmo, K. (2018). Ectoparasite prevalence in Pashmina goats in Changthang: A pastoral nomadic area of Ladakh. Indian Journal of Animal Sciences, 88(3), 281-284.

Shumaker, A. K. (2019). Diagnosis and treatment of canine acral lick dermatitis. Veterinary Clinics of North America - Small Animal Practice, 49(1), 105-123. doi:10.1016/j.cvsm.2018.08.010

Šlapeta, J., Lawrence, A., & Reichel, M. P. (2018). Cat fleas (*Ctenocephalides felis*) carrying *Rickettsia felis* and *Bartonella* species in Hong Kong. Parasitology International, 67(2), 209-212. doi:10.1016/j.parint.2017.12.001

Song, S., Chen, C., Yang, M., Zhao, S., Wang, B., Hornok, S., . . . Wang, Y. (2018). Diversity of *Rickettsia* species in border regions of northwestern China. Parasites and Vectors, 11(1) doi:10.1186/s13071-018-3233-6

Song, N., Lin, A., & Zhao, X. (2018). Insight into higher-level phylogeny of Neuropterida: Evidence from secondary structures of mitochondrial rRNA genes and mitogenomic data. *PloS one*, *13*(1), e0191826.

Soundararajan, C., Nagarajan, K., & Prakash, M. A. (2018). Occurrence of flea infestation on goats under stall fed condition and its control. Journal of Parasitic Diseases, 42(3), 444-448. doi:10.1007/s12639-018-1006-y

Spyrou, M. A., Tukhbatova, R. I., Wang, C. -., Valtueña, A. A., Lankapalli, A. K., Kondrashin, V. V., . . . Krause, J. (2018). Analysis of 3800-year-old *Yersinia pestis* genomes suggests bronze age origin for bubonic plague. Nature Communications, 9(1) doi:10.1038/s41467-018-04550-9

Stenos, J., & Graves, S. R. (2018). Laboratory diagnosis of human infections transmitted by ticks, fleas, mites and lice in Australia. Microbiology Australia, 39(4), 182-184. doi:10.1071/MA18059

Stern, R. M., Luskin, M. R., Clark, R. P., Miller, A. L., & Loscalzo, J. (2018). A headache of a diagnosis. New England Journal of Medicine, 379(5), 475-479. doi:10.1056/NEJMcps1803584

Sudayasa, I. P., Arimaswati, A., Abdullah, M., & Masumi, A. R. (2018). The infuence of head lice (*Pediculus humanus capitis*) infestation to nutritional status and anemia occurence on female elementary school students. Paper presented at the IOP Conference Series: Materials Science and Engineering, , 434(1) doi:10.1088/1757-899X/434/1/012317

Suntsov, V. V. (2018). Quantum speciation of *Yersinia pestis* plague microbe in a heteroimmune environment: In the populations of hibernating Tarbagan marmots (*Marmota sibirica*). Contemporary Problems of Ecology, 11(4), 343-354. doi:10.1134/S199542551804008X

Surkova, E. N., Korallo-Vinarskaya, N. P., Vinarski, M. V., Stanko, M., Warburton, E. M., van der Mescht, L., . . . Krasnov, B. R. (2018). Sexual size dimorphism and sex ratio in arthropod ectoparasites: Contrasting patterns at different hierarchical scales. International Journal for Parasitology, 48(12), 969-978. doi:10.1016/j.ijpara.2018.05.006

Surkova, E. N., Korallo-Vinarskaya, N. P., Vinarski, M. V., van der Mescht, L., Warburton, E. M., Khokhlova, I. S., & Krasnov, B. R. (2018). Body size distribution in flea communities harboured by Siberian small mammals as affected by host species, host sex and scale: Scale matters the most. Evolutionary Ecology, 32(6), 643-662. doi:10.1007/s10682-018-9955-2

Surkova, E. N., Warburton, E. M., van der Mescht, L., Khokhlova, I. S., & Krasnov, B. R. (2018). Body size and ecological traits in fleas parasitic on small mammals in the Palearctic: Larger species attain higher abundance. Oecologia, 188(2), 559-569. doi:10.1007/s00442-018-4235-y

Swales, N., Foster, A., Blackwell, E., & Barnard, N. (2018). Contributing to study of cats with allergic skin disease. Veterinary Record, 183(5), 165. doi:10.1136/vr.k3320

Takahashi, T. (2018). My thoughts on "Circus by Japanese carpenter bees and fleas". Journal of the Japan Research Association for Textile End-Uses, 59(1), 48.

Tan, L., Guan, X., Zhang, L., Zhu, F., & Lei, C. (2018). The complete mitochondrial genome of the flea *Ceratophyllus wui* (Siphonaptera: Ceratophyllidae). Mitochondrial DNA Part B: Resources, 3(1), 401-402. doi:10.1080/23802359.2018.1456370

Teoh, Y. T., Hii, S. F., Graves, S., Rees, R., Stenos, J., & Traub, R. J. (2018). The epidemiology of *Rickettsia felis* infecting fleas of companion animals in eastern Australia. Parasites and Vectors, 11(1) doi:10.1186/s13071-018-2737-4

Ting-Yang, L., Shang-Yi, L., & Tun-Chieh, C. (2018). Murine typhus associated abducens nerve palsy. Kaohsiung Journal of Medical Sciences, 34(9), 537-538. doi:10.1016/j.kjms.2018.02.006

Tokmakova, E. G., Bazanova, L. P., & Voronova, G. A. (2018). The effect of initial infection of *Citellophilus tesquorum altaicus* Ioff, 1936 (Siphonaptera: Ceratophyllidae) with *Yersinia pestis* on its alimentary activity, mortality, and biofilm formation. Entomological Review, 98(4), 414-419. doi:10.1134/S0013873818040036

Tomassone, L., Berriatua, E., De Sousa, R., Duscher, G., Mihalca, A. D., Silaghi, C., . . . Zintl, A. (2018). Neglected vector-borne zoonoses in Europe: Into the wild. Veterinary Parasitology, 251, 17-26. doi:10.1016/j.vetpar.2017.12.018

Toutain, C. E., Seewald, W., & Jung, M. (2018). Pharmacokinetics of lotilaner following a single oral or intravenous administration in cats. Parasites and Vectors, 11(1) doi:10.1186/s13071-018-2966-6

Tyler, S., Roberts, C., Foster, A., Barnard, N., & Murray, J. K. (2018). Owner-reported flea treatment measures and skin disease in cats. Journal of Feline Medicine and Surgery, doi:10.1177/1098612X18773911

Ural, K., Erdoğan, H., & Gültekin, M. (2018). Allergen specific IgE determination by in vitro allergy test in head and facial feline dermatitis: A pilot study. [Baş ve yüz dermatitli kedilerde in vitro alerji testi ile alerjen spesifik IgE tayini: Bir pilot çalışma] Ankara Universitesi Veteriner Fakultesi Dergisi, 65(4), 379-386. doi:10.1501/Vetfak_0000002871

Ursich, E., Anderson, N., Evans, G., & Pearson, R. (2018). Protect yourself, your staff and your clients. Veterinary Record, 182(7), 201. doi:10.1136/vr.k688

van der Mescht, L., Khokhlova, I. S., Warburton, E. M., Dlugosz, E. M., Kotler, B. P., & Krasnov, B. R. (2018). Can we predict the success of a parasite to colonise an invasive host? Parasitology Research, 117(7), 2305-2314. doi:10.1007/s00436-018-5921-8 van der Mescht, L., Warburton, E. M., Khokhlova, I. S., Stanko, M., Vinarski, M. V., Korallo-Vinarskaya, N. P., & Krasnov, B. R. (2018). Biogeography of parasite abundance: Latitudinal gradient and distance decay of similarity in the abundance of fleas and mites, parasitic on small mammals in the palearctic, at three spatial scales. International Journal for Parasitology, 48(11), 857-866. doi:10.1016/j.ijpara.2018.04.005

van Heugten, J., & Wolters, S. (2018). Obituary for a flea. Paper presented at the Springer Proceedings in Mathematics and Statistics, , 261 331-360. doi:10.1007/978-981-13-2487-1_13.

Vasconcelos, E. J. R., Billeter, S. A., Jett, L. A., Meinersmann, R. J., Barr, M. C., Diniz, P. P. V. P., & Oakley, B. B. (2018). Assessing cat flea microbiomes in northern and southern California by 16S rRNA next-generation sequencing. Vector-Borne and Zoonotic Diseases, 18(9), 491-499. doi:10.1089/vbz.2018.2282

Vatta, A. F., Myers, M. R., Rugg, J. J., Chapin, S., Pullins, A., King, V. L., & Rugg, D. (2018). Efficacy and safety of a combination of selamectin plus sarolaner for the treatment and prevention of flea infestations and the treatment of ear mites in cats presented as veterinary patients in the United States. Veterinary Parasitology, doi:10.1016/j.vetpar.2018.11.009

Víchová, B., Bona, M., Miterpáková, M., Kraljik, J., Cabanová, V., Nemčíková, G., . . . Oravec, M. (2018). Fleas and ticks of red foxes as vectors of canine bacterial and parasitic pathogens, in Slovakia, central Europe. Vector-Borne and Zoonotic Diseases, 18(11), 611-619. doi:10.1089/vbz.2018.2314

Vidémont-Drevon, E. (2018). Atopie dermatitis in a Staffordshire bull terrier. [Dermatite atopique chez un staffordshire bull terrier] Point Veterinaire, 49(386), 46-50.

Vieira, F. T., Labruna, M. B., Barbosa, A. C. M. S., Aguiar, A. R., Acosta, I. C. L., Martins, T. F., . . . Braga, F. R. (2018). Occurrence of ticks in dogs in a hospital population in the state of Espírito Santo, Brazil. Pesquisa Veterinaria Brasileira, 38(3), 519-521. doi:10.1590/1678-5150-PVB-5112

Wessels, C., Matthee, S., Espinaze, M. P. A., & Matthee, C. A. (2018). Comparative mtDNA phylogeographic patterns reveal marked differences in population genetic structure between generalist and specialist ectoparasites of the African penguin (*Spheniscus demersus*). Parasitology Research, doi:10.1007/s00436-018-6150-x

Wheeler, D. W., Trout, C. M., Thompson, C. M., Winkle, J. R., & White, W. H. (2018). Evaluation of an 11.2% spinetoram topical spot-on solution for the control of experimental and natural flea (*Ctenocephalides felis*) infestations on cats in Europe. Veterinary Parasitology, 258, 99-107. doi:10.1016/j.vetpar.2018.05.018

Wiese, S., Elson, L., & Feldmeier, H. (2018). Tungiasis-related life quality impairment in children living in rural Kenya. PLoS Neglected Tropical Diseases, 12(1) doi:10.1371/journal.pntd.0005939

Wilson, J. M., & Daniel, M. (2018). Historical reconstruction of the community response, and related epidemiology, of a suspected biological weapon attack in Ningbo, China (1940). Intelligence and National Security, doi:10.1080/02684527.2018.1536351

Wismer, T., & Means, C. (2018). Toxicology of newer insecticides in small animals. Veterinary Clinics of North America - Small Animal Practice, 48(6), 1013-1026. doi:10.1016/j.cvsm.2018.06.005

Wright, I. (2018). Lotilaner - A novel formulation for cats provides systemic tick and flea control. Parasites and Vectors, 11(1) doi:10.1186/s13071-018-2970-x

Yáñez-Meza, A., Moreno, L., & Botto-Mahan, C. (2018). Ectoparásitos del roedor endémico *Abrocoma bennetti* (Hystricomorpha: Abrocomidae) en Chile semiárido. *Gayana (Concepción)*, *82*(1), 94-97.

Young, T. P., Porensky, L. M., Riginos, C., Veblen, K. E., Odadi, W. O., Kimuyu, D. M., . . . Young, H. S. (2018). Relationships between cattle and biodiversity in multiuse landscape revealed by Kenya long-term exclosure experiment. Rangeland Ecology and Management, 71(3), 281-291. doi:10.1016/j.rama.2018.01.005

Yousefi, A., Rahbari, S., & Eslami, A. (2018). Ectoparasites associated with small mammals (orders Insectivora, Eulipotyphla, and Rodentia) in Razan Plain, western region of Iran. Comparative Clinical Pathology, 27(3), 667-671. doi:10.1007/s00580-018-2647-z

Yudina, M. A., Bykov, R. A., Kotti, B. K., Vysochina, N. P., Stakheev, V. V., Broshkov, A. D., . . . Ilinsky, Y. Y. (2018). *Wolbachia* infection in flea populations (Insecta: Siphonaptera). Zhurnal Obshchei Biologii, 79(3), 237-246. doi:10.7868/S0044459618030053

Yue, R. P. H., & Lee, H. F. (2018). Pre-industrial plague transmission is mediated by the synergistic effect of temperature and aridity index. BMC Infectious Diseases, 18(1) doi:10.1186/s12879-018-3045-5

Zapata-Valdés, C., Avila-Flores, R., Gage, K., Holmes, J., Montenierri, J., Kosoy, M., & Suzán, G. (2018). Mammalian hosts, vectors, and bacterial pathogens in communities with and without black-tailed prairie dogs (*Cynomys ludovicianus*) in northwestern Mexico. Journal of Wildlife Diseases, 54(1), 26-33. doi:10.7589/2016-09-214

Zeppelini, C. G., DE ALMEIDA, A. M. P., & Cordeiro-Estrela, P. (2018). Ongoing quiescence in the Borborema Plateau plague focus (Paraiba, Brazil). Anais Da Academia Brasileira De Ciencias, 90(3), 3007-3015. doi:10.1590/0001-3765201820170977

Zhao, F., Zhang, T., Su, J., Huang, Z., Wu, A., & Lin, G. (2018). Genetic differentiation of the oriental rat flea, *Xenopsylla cheopis*, from two sympatric host species. Parasites and Vectors, 11(1) doi:10.1186/s13071-018-2903-8

Zurita, A., Callejón, R., De Rojas, M., & Cutillas, C. (2018). Morphological and molecular study of the genus *Nosopsyllus* (Siphonaptera: Ceratophyllidae). *Nosopsyllus barbarus* (Jordan & Rothschild 1912) as a junior synonym of *Nosopsyllus fasciatus* (Bosc d'antic, 1800). Insect Systematics and Evolution, 49(1), 81-101. doi:10.1163/1876312X-00002164

Zurita, A., Djeghar, R., Callejón, R., Cutillas, C., Parola, P., & Laroche, M. (2018). Matrix-assisted laser desorption/ionization time-of-flight mass spectrometry as a useful tool for the rapid identification of wild flea vectors preserved in alcohol. Medical and Veterinary Entomology, doi:10.1111/mve.12351

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Publications on Siphonaptera of Allen H. Benton

(courtesy T. Galloway)

Benton, A.H. 1955. The taxonomy and biology of *Epitedia wenmanni* (Rothschild, 1904) and *E. testor* (Rothschild, 1915) (Hystrichopsyllidae: Siphonaptera). Journal of Parasitology, 41: 491-495.

Benton, A.H. 1955. A modified technique for preparing whole mounts of Siphonaptera. Journal of Parasitology, 41: 422-323

Benton, A.H. 1955. Notes on some rare or little-known New York Siphonaptera. Journal of the New York Entomological Society, 63: 139-140.

Benton, A.H., and R.F. Krug. 1956. Mammals and siphonapterous parasites of Rensselaer County, New York. New York State Museum and Science Service, Bulletin No. 353, Albany, New York. pp. 3-22.

Benton, A.H., R. Cerwonka, and J. Hill. 1959. Observations on host perception in fleas. Journal of Parasitology, 45: 614.

Benton, A.H., and R.H. Cerwonka. 1960. Host relationships of some eastern Siphonaptera. The American Midland Naturalist, 63: 383-391.

Benton, A.H., and H.E. Shadowen. 1960. Occurrence of *Pulex simulans* Baker in Louisiana. The Southwestern Naturalist, 5: 104.

Benton, A.H., and V. Shatrau. 1960. Bird fleas in New York State. The Kingbird, 9: 6-7.

Benton, A.H., and V. Shatrau. 1962. Notes on the collection of bird fleas from Grand Manan, New Brunswick. The Canadian Entomologist, 94: 743-745.

Benton, A.H., and D. Smiley. 1963. The fleas of Ulster County New York. John Burroughs Natural History Society, Bulletin No. 6 of the Research and Records Committee. 7 pp. + 2 figs.

Benton, A.H., and R.H. Cerwonka. 1963. The Siphonaptera of Whiteface Mountain. Whiteface Mountain Summer Program – 1963, Atmospheric Sciences Research Center, State University of New York. pp. 4-14.

Benton, A.H. 1964. Review of “G.H.E. Hopkins, and M. Rothschild, An Illustrated Catalogue of the Rothschild Collection of Fleas (Siphonaptera) in the British Museum (Natural History). Volume III: Hystrichopsyllidae (Acedestiinae, Anomiopsyllinae, Hystrichopsyllinae, Neopsyllinae, Rhadinopsyllinae, and Stenoponiinae)”. Quarterly Review of Biology, 39: 206.

Benton, A.H., and H.J. Altmann, 1964. A study of fleas found on *Peromyscus* in New York. Journal of Mammalogy, 45: 31-36.

Benton, A.H., and V. Shatrau. 1965. The bird fleas of eastern North America. The Wilson Bulletin, 77: 76-81.

Benton, A.H., and S.Y. Lee. 1965. Sensory reactions of Siphonaptera in relation to host-finding. The American Midland Naturalist, 74: 119-125.

Benton, A.H., and S.Y. Lee. 1965. Host finding reactions in some fleas. p. 792. *In*: XIIth International Congress of Entomology, London, UK, 8-16 July, 1964.

Benton, A.H., and P.F. Connor. 1966. Siphonaptera collected on Tug Hill, Lewis County, New York. Bulletin No. 406, New York State Museum and Science Service. pp. 76-78.

Benton, A.H. 1967. A case of teratology in *Monopsyllus vison* (Baker). Journal of the New York Entomological Society, 75: 31-33.

Benton, A.H. 1967. *Peromyscopsylla hamifer hamifer* (Rothschild): an addition to the entomological fauna of New York. Journal of the New York Entomological Society, 75: 159-160.

Holland, G.P., and A.H. Benton. 1968. Siphonaptera from Pennsylvania mammals. The American Midland Naturalist, 80: 252-261.

Benton, A.H., and D.L. Kelly. 1969. Notes on the biology of *Ctenophthalmus p. pseudagyrtes* Baker in the northeast (Siphonaptera: Hystrichopsyllidae). Journal of the New York Entomological Society, 77: 70-74.

Benton, A.H., H.H., Tucker, Jr., and D.L. Kelly. 1969. Siphonaptera from northern New York. Journal of the New York Entomological Society, 77: 193-198.

Benton, A.H. 1970. An annotated list of the fleas (Siphonaptera) of West Virginia. Proceedings of the West Virginia Academy of Sciences, 42: 35-39.

Benton, A.H., and D.H. Miller. 1970. Ecological factors in the distribution of the flea *Peromyscopsylla hamifer hamifer* (Rothschild). The American Midland Naturalist, 83: 301-303.

Miller, D.H., and A.H. Benton. 1970. Cold tolerance in some adult fleas (Ceratophyllidae: Siphonaptera). The Canadian Field-Naturalist, 84: 396-397.

Benton, A.H., and D. Kelly. 1971. Siphonaptera from Long Island, New York. *In*: P.F. Connor, The mammals of Long Island, New York. New York State Museum and Science Service, Bulletin No. 416. pp. 77-78.

Benton, A.H., O.R. Larson, and B.A. ven Huizen. 1971. Siphonaptera of Itasca State Park region. Journal of the Minnesota Academy of Science, 37: 91-92.

Benton, A.H. 1972. 25. Fleas and the second-order environment. Journal of Medical Entomology, 9: 602.

Miller, D.H., and A.H. Benton. 1973. An annotated list of the Siphonaptera of Connecticut. Journal of the New York Entomological Society, 81: 210-213.

Benton, A.H., and D. Kelly. 1975. An annotated list of New York Siphonaptera. Journal of the New York Entomological Society, 83: 143-156.

Benton, A.H., and R.M. Timm. 1977. Description of the female of *Rhadinopsylla media* (Siphonaptera: Hystrichopsyllidae). Journal of Medical Entomology, 13: 473-475.

Benton, A.H. 1979. Another non-jumper. Flea News, 16: 2. (*Conorhinopsylla stanfordi*)

Benton, A.H., M. Surman, W.L. Krinsky. 1979. Observations on the feeding habits of some larval fleas (Siphonaptera). Journal of Parasitology, 65: 671-672.

Benton, A.H. 1980. A new species of *Nearctopsylla* Rothschild 1915 (Siphonaptera: Hystrichopsyllidae) from the southern Appalachians. Journal of Parasitology, 66: 841-843.

Benton, A.H. 1980. An atlas of the fleas of the eastern United States. Marginal Media, Fredonia, New York. xvi + 177 pp. (This includes The Siphonaptera of Minnesota, pp. 158-177.)

Benton, A.H., and J.F. Day. 1980. Seasonal changes in the flea fauna of nests of the southern flying squirrel. pp. 401-403. *In*: Proceedings of the International Conference on Fleas, *Edited by* R. Traub and H. Starcke. Ashton Wold, Peterborough, UK, 21-25 June, 1977.

Day, J.F., and A.H. Benton. 1980. Population dynamics and coevolution of adult siphonapteran parasites of the southern flying squirrel (*Glaucomys volans volans*). The American Midland Naturalist, 103: 333-338.

Benton, A.H. 1982. Life strategies in the order Siphonaptera: a review of some unsolved problems. Advances in Biosciences, 1: 28-34.

Benton, A.H. 1983. An illustrated key to the fleas of the eastern United States. Marginal Media, Fredonia, New York. Bioguide No.3. 34 pp.

Benton, A.H., M. Surman, and R.E. Elbel. 1983. Comparative studies of the larvae of fleas of the flying squirrel, *Glaucomys v. volans*. Proceedings of the 38th International Conference on Diseases Communicable to Man. One page, no pagination.

Benton, A.H. 1985. Fleas of medical and veterinary importance. Marginal Media, Fredonia, New York. 8 pp.

Galloway, T.D., G.E. Haas, and A.H. Benton. 1989. Corrections and additions to G.P. Holland's "Fleas of Canada, Alaska and Greenland (Siphonaptera)". The Canadian Entomologist, 121: 609-618.

Benton, A.H. 1990. A Siphonapterous Sonnet. Marginal Media, Fredonia, New York. p. 50. (Published under the pseudonym, Albert E. Fitzwarren.)

Linley, J.R., A.H. Benton, and J.F. Day. 1994. Ultrastructure of the eggs of seven species of fleas (Siphonaptera). Journal of Medical Entomology, 31: 813-827.

Benton, A.H. 1998. Hybridization in North American bird fleas of the genus *Ceratophyllus* (Siphonaptera: Ceratophyllidae). Journal of Medical Entomology, 35: 433-434.

Glorious nature
Fills each day with such beauty –
How can I leave it?

Haiku by Allen H. Benton (1990) *In*: The Wheel of Life Haiku by Followers of Basho. Nymphaea Productions.